amylase production from bacteria

nmz

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1 Abstract

This study presents a systematic investigation of bacterial α-amylase catalytic properties through integrated spectrophotometric, kinetic, and response surface methodologies [@ataallahiSimple2stepPurification2021]. The analytical workflow encompassed glucose calibration via 3,5-dinitrosalicylic acid (DNS) assay, fermentation kinetics under varying salt concentrations, response surface optimization of pH and temperature, comparative substrate specificity across four botanical starches, and Michaelis-Menten parameter estimation [@ataallahiSimple2stepPurification2021;

@aENZYMICSTUDIESAlphaAMYLASE2019;@ataallahiSimple2stepPurification2021b]. The glucose standard curve demonstrated exceptional linearity (R^2 = 0.9909), establishing a robust empirical transfer function for reducing sugar quantification. Response surface methodology identified optimal catalytic conditions at pH 5.96 and 49.89°C, with both eigenvalues negative (λ_1 = -0.381, λ_2 = -0.471), confirming a true global maximum. Kinetic analysis revealed pronounced substrate-dependent variation in catalytic efficiency, with soluble starch exhibiting the highest V_{max} (0.139 ΔA_{620} min^{-1}) and lowest K_m (3.835

2 Data Import

- load the libraries
- load the data for standard curve
- load the experimental data

```
# loading libraries
library(ggpubr)
library(tidyverse)
library(readxl)
library(Myfuns)
library(patchwork)
library(broom)

# loading standard curve data
std_c <- read_excel("../Data/Salting out and DNS assay.xlsx",
  sheet = "Sheet3"
)
names(std_c)
```

```
[1] "Glu_conc" "Abs"
```

```
# loading Experiment Data
so <- read_excel("../Data/Salting out and DNS assay.xlsx",
  sheet = "Sheet2"
)
names(so)
```

```
[1] "Time"      "Salt_conc" "Abs"
```

```
buf <- 0.6713
so2 <- so %>% mutate(Abs = Abs - buf)
```

3 Relationship of the Glucose concentration and Absorbance

- fit the linear regression
- visualize the linear relationship

```
std_c_eq1 <- lm(Glu_conc ~ Abs, std_c)
summary(std_c_eq1)
```

```

Call:
lm(formula = Glu_conc ~ Abs, data = std_c)

Residuals:
      Min        1Q    Median        3Q       Max
-0.041356 -0.027569 -0.006494  0.006797  0.065767

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.03242    0.01513  -2.143   0.0478 *
Abs          0.46291    0.01109  41.760   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.03454 on 16 degrees of freedom
Multiple R-squared:  0.9909,    Adjusted R-squared:  0.9903
F-statistic:  1744 on 1 and 16 DF,  p-value: < 2.2e-16
```

```
std_c_eq2 <- lm(Abs ~ Glu_conc, std_c)
summary(std_c_eq2)
```

```

Call:
lm(formula = Abs ~ Glu_conc, data = std_c)

Residuals:
     Min       1Q   Median       3Q      Max
-0.14011 -0.02057  0.01599  0.05024  0.09916

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.07986    0.03104   2.573   0.0204 *
Glu_conc     2.14059    0.05126  41.760   <2e-16 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

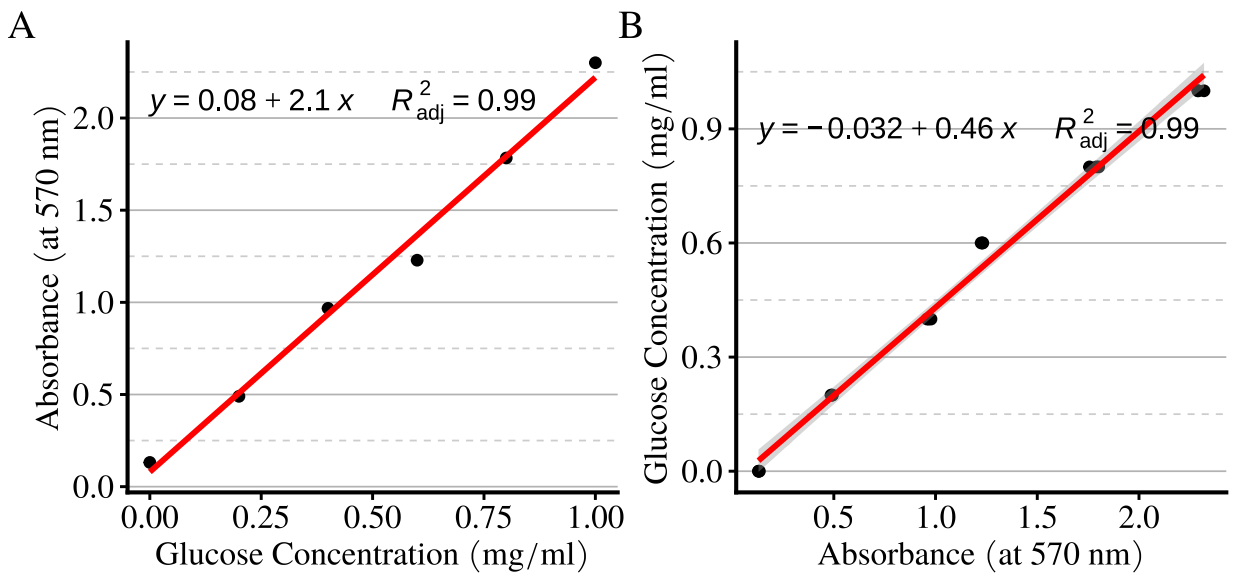
Residual standard error: 0.07428 on 16 degrees of freedom
Multiple R-squared:  0.9909,    Adjusted R-squared:  0.9903
F-statistic:  1744 on 1 and 16 DF,  p-value: < 2.2e-16
```

```
p1 <- std_c %>% reframe(Abs=mean(Abs),.by = Glu_conc) %>%
  ggplot(aes(Glu_conc, Abs)) +
  geom_point() +
  geom_smooth(method = "lm", color = "red", se = F) +
  # geom_smooth(linetype = "dashed", color = "blue") +
  stat_regline_equation(
    aes(label = paste(..eq.label..,
                      ..adj.rr.label..,
                      sep = "~~~~~")),
    decreasing = FALSE
  )+
  labs(x = "Glucose Concentration (mg/ml)",
       y = "Absorbance (at 570 nm)")+
  Mytheme(font = "TeX Gyre Termes")


p2 <- std_c %>% ggplot(aes(Abs, Glu_conc)) +
  geom_point() +
  geom_smooth(method = "lm", color = "red") +
  # geom_smooth(linetype = "dashed", color = "blue") +
  stat_regline_equation(
    aes(label = paste(..eq.label..,
                      ..adj.rr.label..,
                      sep = "~~~~~")),
    # aes(label = paste0("atop(", ..eq.label.., ",", ..adj.rr.label.., ")")),
    decreasing = FALSE
  )+
  labs(x= "Absorbance (at 570 nm)",
       y= "Glucose Concentration (mg/ml)")+
  Mytheme(font = "TeX Gyre Termes")

p1+p2+
  plot_annotation(tag_levels = "A")
```

```
`geom_smooth()` using formula = 'y ~ x'
`geom_smooth()` using formula = 'y ~ x'
```



4 Prepare the experiment data for analysis

```
names(so)
```

```
[1] "Time"      "Salt_conc" "Abs"
```

```
buf <- 0.6713
so2 <- so %>% mutate(Abs = Abs - buf)

pre_glu <- predict(std_c_eq1, data.frame(Abs = so2$Abs))
# pre_glu2 <- predict(std_c_eq2, data.frame(Abs = so2$Abs))

so2$glu <- pre_glu
so2 <- so2[-c(1:6), ]
p3 <- so2 %>%
  mutate(Salt_conc = factor(Salt_conc)) %>%
  ggplot(aes(Time, glu, group = factor(Salt_conc))) +
  geom_point(aes(color = Salt_conc)) +
  geom_smooth(aes(color = Salt_conc),
    se = F,
    # method = "lm",
    # formula = y ~ poly(x, 2, raw = T)
  ) +
  labs(
    x = "Growth time (hour)",
    y = "Glucose Concentration
    (mg/ml)",
```

```
    color = "Salt Concentration"
  ) +
  scale_color_discrete(
    breaks = c(10, 20, 30),
    labels = c("10 %", "20 %", "30 %")
  ) +
  theme_minimal() +
  theme(
    text = element_text(
      family = mfont,
      color = "black",
      size = 12
    ),
    axis.line = element_line(color = "black"),
    axis.text = element_text(color = "black"),
    legend.position = "bottom",
    legend.margin = margin(-10, 0, 0, 0),
    plot.margin = margin(5, 5, 5, 5)
  )

## For Contour plot
salt_range <- range(so2$Salt_conc)
time_range <- range(so2$Time)

salt_seq <- seq(salt_range[1], salt_range[2], length.out = 30)
time_seq <- seq(time_range[1], time_range[2], length.out = 30)
prediction_grid <- expand.grid(Salt_conc = salt_seq, Time = time_seq)

prediction_grid$glu <- predict(lm(glu ~ poly(Salt_conc, 2, raw = T) *
                                    poly(Time, 2, raw = T), data = so2),
  newdata = prediction_grid
)

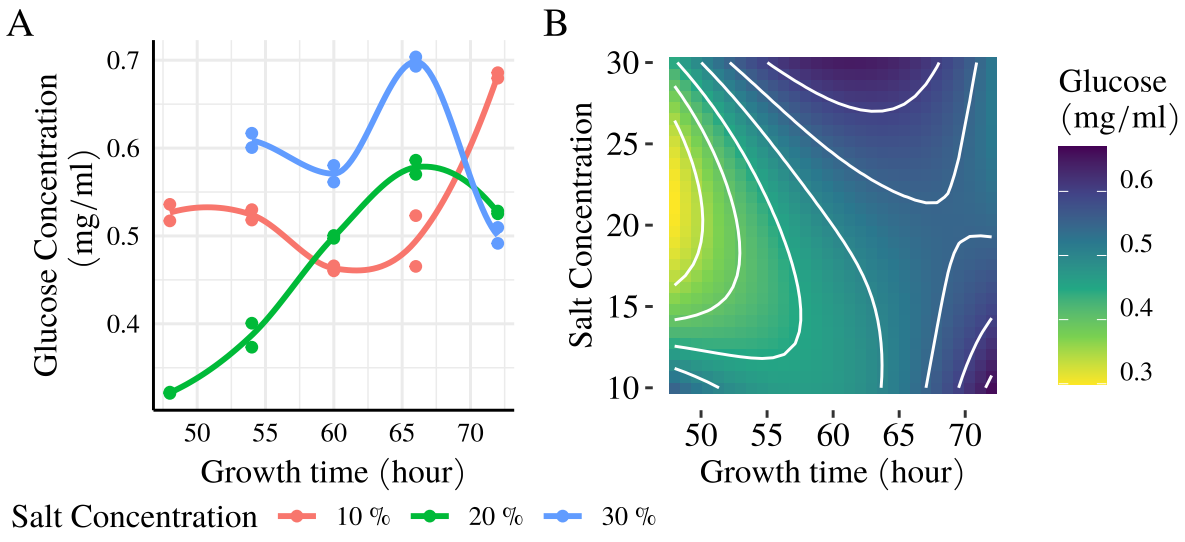
p4 <- prediction_grid %>%
  ggplot(aes(Time, Salt_conc)) +
  geom_tile(aes(fill = glu)) +
  geom_contour(aes(z = glu), color = "white") +
  scale_fill_viridis_c(
    name = "Glucose\n(mg/ml)", direction = -1,
    limits = c(0.3, 0.67),
    breaks = seq(0.3, 0.67, 0.1)
  ) +
  labs(
    x = "Growth time (hour)",
```

```
    y = "Salt Concentration"

  ) +
  theme(
    text = element_text(
      family = mfont,
      size = 12,
      color = "black"
    ),
    axis.text = element_text(
      family = mfont,
      size = 12,
      color = "black"
    ),
    axis.line = element_blank(),
    legend.spacing.x = unit(5, "pt"),
    legend.text = element_text(
      margin = margin(l = 5),
      vjust = -.2
    ),
    panel.background = element_blank()
  )

p3+p4+plot_annotation(tag_levels = "A")
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



4.1 model used for contour plot

```
mod_used <- lm(glu ~ poly(Salt_conc, 2, raw = T) * poly(Time, 2, raw = T),
               data = so2)
summary(mod_used)
```

```
Call:
lm(formula = glu ~ poly(Salt_conc, 2, raw = T) * poly(Time, 2,
    raw = T), data = so2)

Residuals:
      Min        1Q    Median        3Q       Max
-0.083070 -0.019811 -0.003844  0.024736  0.078950

Coefficients:
                                                    Estimate Std. Error
(Intercept)                                         1.317e+01  4.184e+00
poly(Salt_conc, 2, raw = T)1                       -1.161e+00  5.006e-01
poly(Salt_conc, 2, raw = T)2                        2.013e-02  1.351e-02
poly(Time, 2, raw = T)1                            -4.081e-01  1.401e-01
poly(Time, 2, raw = T)2                             3.296e-03  1.156e-03
poly(Salt_conc, 2, raw = T)1:poly(Time, 2, raw = T)1  3.592e-02  1.666e-02
poly(Salt_conc, 2, raw = T)2:poly(Time, 2, raw = T)1 -5.921e-04  4.458e-04
poly(Salt_conc, 2, raw = T)1:poly(Time, 2, raw = T)2 -2.799e-04  1.368e-04
poly(Salt_conc, 2, raw = T)2:poly(Time, 2, raw = T)2  4.436e-06  3.634e-06
                                                    t value Pr(>|t|)
(Intercept)                                           3.149  0.00529 **
poly(Salt_conc, 2, raw = T)1                         -2.320  0.03163 *
poly(Salt_conc, 2, raw = T)2                          1.490  0.15269
poly(Time, 2, raw = T)1                              -2.913  0.00892 **
poly(Time, 2, raw = T)2                               2.851  0.01022 *
poly(Salt_conc, 2, raw = T)1:poly(Time, 2, raw = T)1  2.155  0.04416 *
poly(Salt_conc, 2, raw = T)2:poly(Time, 2, raw = T)1 -1.328  0.19988
poly(Salt_conc, 2, raw = T)1:poly(Time, 2, raw = T)2 -2.046  0.05487 .
poly(Salt_conc, 2, raw = T)2:poly(Time, 2, raw = T)2  1.221  0.23719
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.04749 on 19 degrees of freedom
Multiple R-squared:  0.841, Adjusted R-squared:  0.774
F-statistic: 12.56 on 8 and 19 DF,  p-value: 4.099e-06
```

5 Optimization of pH and Temperature for maximum enzyme activity

- load data
- fit to response surface
- visualize the response surface
- calculation of optimal condition ## load library and data

```
library(rsm)
RSM <- read_excel("../Data/(26.06.2026) absorbance data.xlsx")
RSM2 <- RSM %>% mutate(Temp = str_remove(Temp, "°C") %>% as.numeric())

codedrsm2 <- coded.data(RSM2,
                        x1~(Temp -50)/10,
                        x2 ~(pH - 6)/0.5)
```

5.1 fit the model

```
rsmmod <- rsm(avg~SO(x1, x2), codedrsm2)
print(rsmmod)
```

```
Call:
rsm(formula = avg ~ SO(x1, x2), data = codedrsm2)

Coefficients:
   (Intercept)    FO(x1, x2)x1    FO(x1, x2)x2     TWI(x1, x2)  PQ(x1, x2)x1^2
       1.22668         -0.01519         -0.05774         -0.06495         -0.45733
PQ(x1, x2)x2^2
      -0.39470
```

```
summary(rsmmod)
```

```
Call:
rsm(formula = avg ~ SO(x1, x2), data = codedrsm2)

             Estimate Std. Error  t value  Pr(>|t|)
(Intercept)  1.226682   0.019478  62.9775 1.078e-09 ***
x1          -0.015189   0.013642  -1.1134  0.308139
x2          -0.057740   0.013851  -4.1686  0.005888 **
x1:x2       -0.064950   0.019491  -3.3324  0.015760 *
x1^2        -0.457332   0.014798 -30.9044 7.622e-08 ***
x2^2        -0.394703   0.015681 -25.1709 2.589e-07 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Multiple R-squared:  0.9956,    Adjusted R-squared:  0.9919
F-statistic: 270.6 on 5 and 6 DF,  p-value: 5.62e-07

Analysis of Variance Table

Response: avg
            Df  Sum Sq Mean Sq  F value    Pr(>F)
FO(x1, x2)   2 0.02779 0.01389   9.1435   0.01508
TWI(x1, x2)  1 0.01687 0.01687  11.1048   0.01576
PQ(x1, x2)   2 2.01124 1.00562 661.8022 9.189e-08
Residuals    6 0.00912 0.00152
Lack of fit  3 0.00261 0.00087   0.4017   0.76322
Pure error   3 0.00650 0.00217

Stationary point of response surface:
         x1          x2
-0.01147885 -0.07219898

Stationary point in original units:
     Temp        pH
49.885212  5.963901

Eigenanalysis:
eigen() decomposition
$values
[1] -0.3809044 -0.4711314

$vectors
         [,1]       [,2]
x1  0.3910698 -0.9203610
x2 -0.9203610 -0.3910698
```

```
canonical(rsmmod)
```

```
$xs
         x1          x2
-0.01147885 -0.07219898

$eigen
eigen() decomposition
$values
[1] -0.3809044 -0.4711314
```

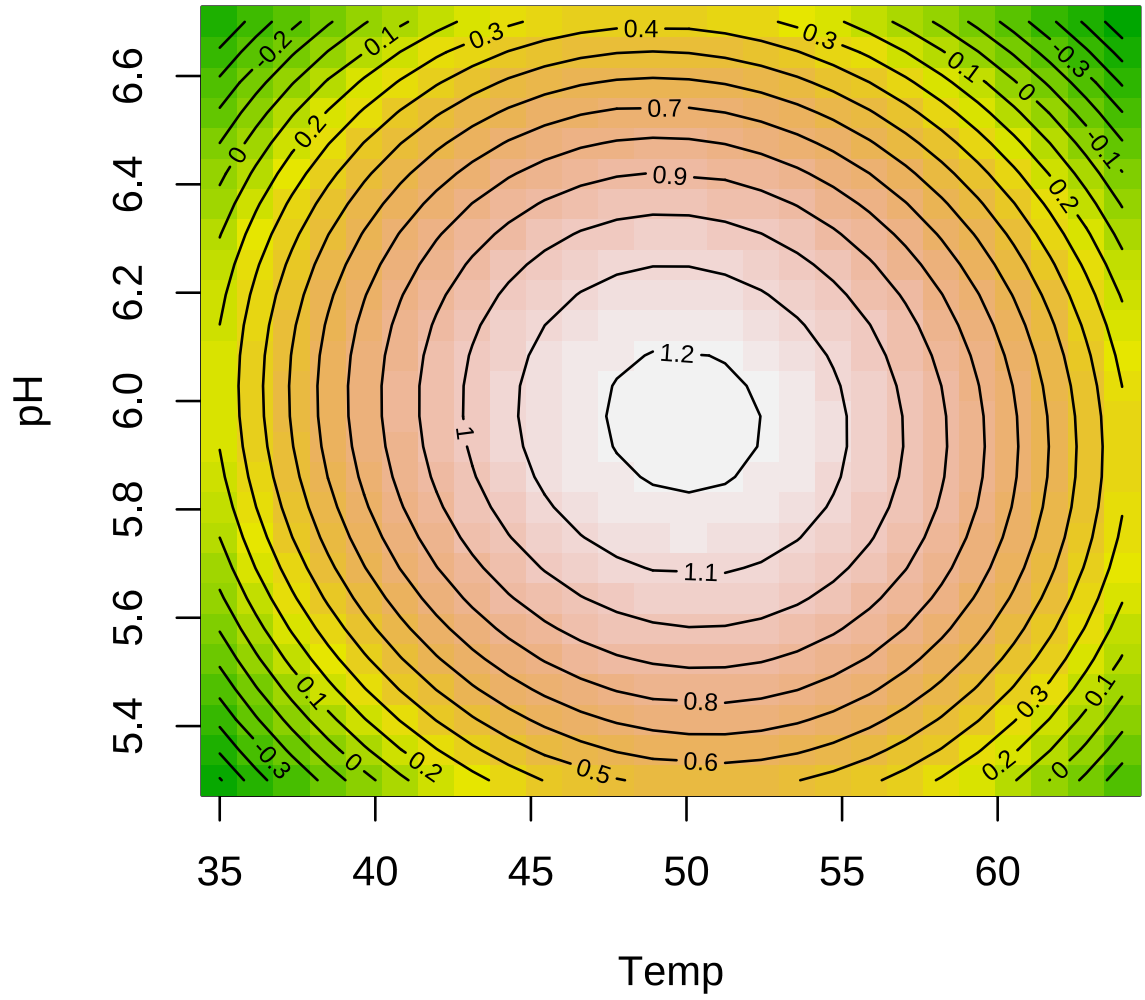
```
$vectors
         [,1]       [,2]
x1  0.3910698 -0.9203610
x2 -0.9203610 -0.3910698
```

```r
canonical.path(rsmmod)
```

```
   dist     x1     x2 |  Temp     pH |   yhat
1  -5.0 -1.967  4.530 | 30.33 8.2650 | -8.295
2  -4.5 -1.771  4.069 | 32.29 8.0345 | -6.483
3  -4.0 -1.576  3.609 | 34.24 7.8045 | -4.865
4  -3.5 -1.380  3.149 | 36.20 7.5745 | -3.437
5  -3.0 -1.185  2.689 | 38.15 7.3445 | -2.200
6  -2.5 -0.989  2.229 | 40.11 7.1145 | -1.152
7  -2.0 -0.794  1.769 | 42.06 6.8845 | -0.296
8  -1.5 -0.598  1.308 | 44.02 6.6540 |  0.372
9  -1.0 -0.403  0.848 | 45.97 6.4240 |  0.848
10 -0.5 -0.207  0.388 | 47.93 6.1940 |  1.134
11  0.0 -0.011 -0.072 | 49.89 5.9640 |  1.229
12  0.5  0.184 -0.532 | 51.84 5.7340 |  1.134
13  1.0  0.380 -0.993 | 53.80 5.5035 |  0.848
14  1.5  0.575 -1.453 | 55.75 5.2735 |  0.372
15  2.0  0.771 -1.913 | 57.71 5.0435 | -0.295
16  2.5  0.966 -2.373 | 59.66 4.8135 | -1.151
17  3.0  1.162 -2.833 | 61.62 4.5835 | -2.199
18  3.5  1.357 -3.293 | 63.57 4.3535 | -3.436
19  4.0  1.553 -3.754 | 65.53 4.1230 | -4.867
20  4.5  1.748 -4.214 | 67.48 3.8930 | -6.485
21  5.0  1.944 -4.674 | 69.44 3.6630 | -8.294
```

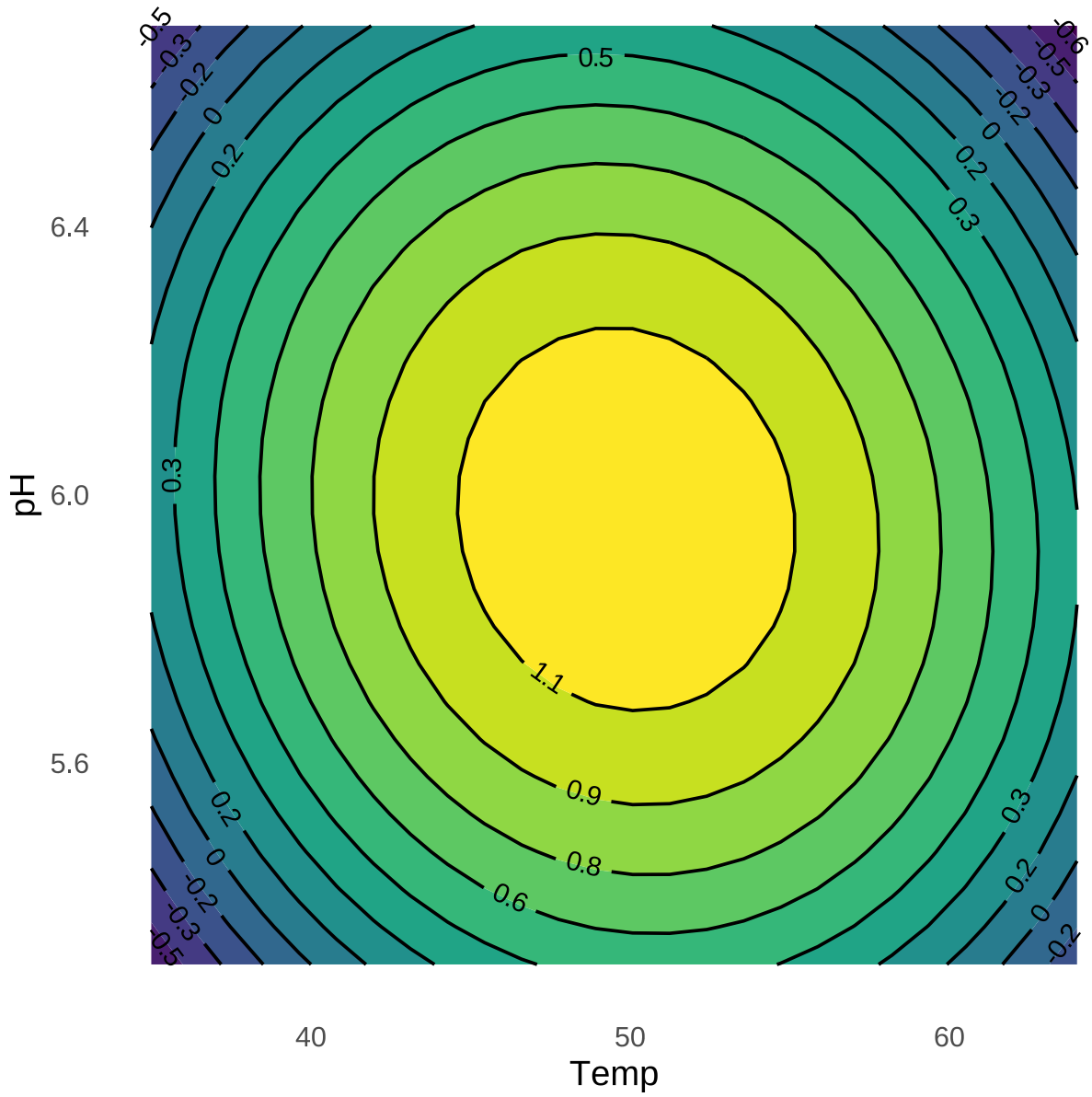
5.2 visualize

```r
contour(rsmmod, ~ x1 + x2,
        at = canonical(rsmmod)$xs,
        image = T, nlevels = 15, decode = T
)
```

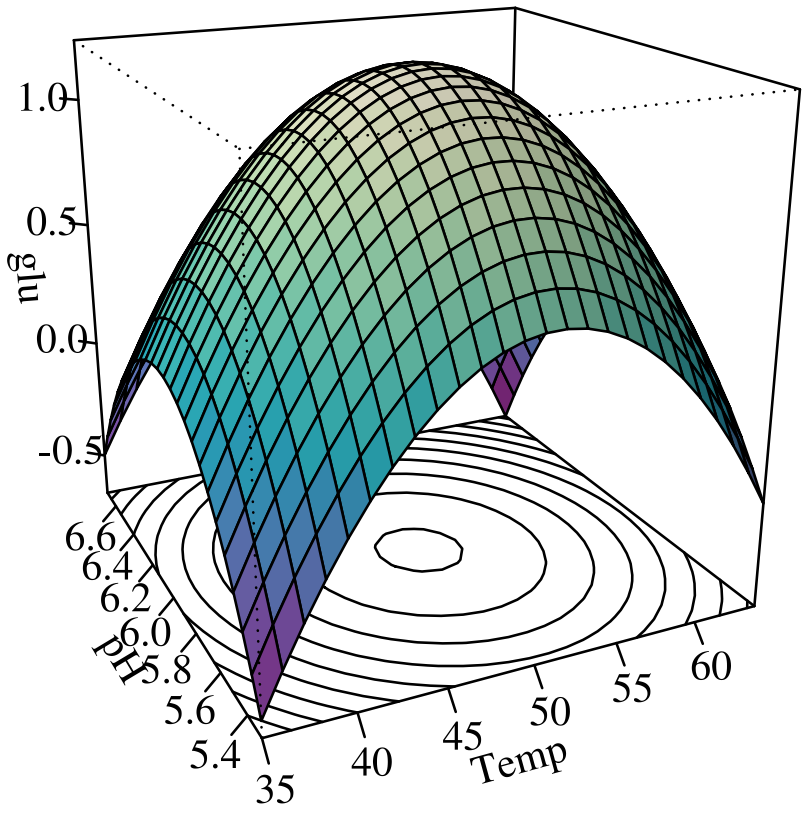



```
# from Myfuns library
my_gg_rsm(rsmmod, ~ x1 + x2,
          at = canonical(rsmmod)$xs,
          color = 'black',
          text_size = 3,
          decode =T)
```

```
[[1]]
```




```
persp(rsmmod, ~ x1 + x2,
      at = canonical(rsmmod)$xs,
      contours = list(z = "bottom"), ticktype = "detailed", nticks = 7,
      shade = .2,
      col = hcl.colors(20, "Purple-Yellow"),
      family = mfont,
      zlab = "glu"
)
```




```
# library(rgl)
# x1_seq <- seq(min(rsmmod$data$x1), max(rsmmod$data$x1), length.out = 30)
# x2_seq <- seq(min(rsmmod$data$x2), max(rsmmod$data$x2), length.out = 30)
# grid <- expand.grid(x1 = x1_seq, x2 = x2_seq)
# grid$z <- predict(rsmmod, newdata = grid)
#
# # Convert to matrix
# z_matrix <- matrix(grid$z, nrow = length(x1_seq), ncol = length(x2_seq))
#
# persp3d(x1_seq, x2_seq, z_matrix,
#   col = hcl.colors(20, "Purple-Yellow")[cut(z_matrix, 20)],
```

```
#   xlab = "Temperature",
#   ylab = "pH",
#   zlab = "Absorbance",
#   ticktype = "detailed"
# )
#
# # Calculate contour lines
# contour_lines <- contourLines(x1_seq, x2_seq, z_matrix)
#
# # Add contours at the BOTTOM
# for (i in seq_along(contour_lines)) {
#   lines3d(
#     x = contour_lines[[i]]$x,
#     y = contour_lines[[i]]$y,
#     z = min(z_matrix),
#     col = "gray50",
#     lwd = 1,
#     alpha = 0.5
#   )
# }
#
# # Add contours ON the SURFACE
# for (i in seq_along(contour_lines)) {
#   z_vals <- rep(contour_lines[[i]]$level, length(contour_lines[[i]]$x))
#   lines3d(
#     x = contour_lines[[i]]$x,
#     y = contour_lines[[i]]$y,
#     z = z_vals,
#     col = "black",
#     lwd = 2
#   )
# }
# # Add canonical point
# canon <- canonical(rsmmod)$xs
# canon_z <- predict(rsmmod, newdata = as.data.frame(t(canon)))
# points3d(canon["x1"], canon["x2"], canon_z,
#   col = "red", size = 10
# )
#
# # Add label for canonical point
# text3d(canon["x1"], canon["x2"], canon_z * 1.1,
#   texts = "Optimum", col = "red"
# )
# axes3d()
```

```
# rgl.snapshot("my_3d_plot.png", fmt = "png")
```

```
# knitr::include_graphics("my_3d_plot.png")
```

5.3 What is optimum condition and its predicted response

```
# optimum condition
code2val(canonical((rsmmod))$xs, codings = codings(codedrsm2))
```

```
     Temp        pH
49.885212  5.963901
```

```
prediction <- predict(rsmmod, newdata  = data.frame(
  x1 = canonical(rsmmod)$xs[1],
  x2 = canonical(rsmmod)$xs[2]),
  interval = "confidence")

#predicted response
kable(prediction)
```

	fit	lwr	upr
x1	1.228853	1.181245	1.276462

6 Amylase activity on four different starch of different concentration

- load and tidy the data

```
UV_04_08_2006_ <- read_excel("../Data/UV (04.08.2006) - completed.xlsx",
                             sheet = "Sheet1")
```

```
New names:
* `` -> `...7`
* `` -> `...8`
* `` -> `...9`
```

```
UV <- UV_04_08_2006_[1:42,1:6]
UV <- UV %>% filter(!is.na(`0 min`))

ns <- names(UV)
ns[1] <- "Starch"
```

```r
names(UV) <- ns

UV <- UV %>% filter(!str_detect(`0 min`, "min"))

starch <- c("Soluble Starch","Topioca Starch","Corn Starch","Rice Starch")
starch <- c(rep(starch, each=2), "Negative control")
starch <- rep(starch, times = 4)
UV[,1] <- starch

lvl <- c(
  "Soluble Starch",
  "Corn Starch",
  "Rice Starch",
  "Topioca Starch",
  "Negative control"
)

UV2 <- UV %>%
  mutate(sub_conc=rep(c(1, 0.5, 1.5, 2), each = 9)) %>%
  pivot_longer(2:6,
                    names_to = "Time",
                    values_to = "ABS") %>%
  mutate(Time=str_remove(Time, " min"),
         Time = as.double(Time)) %>%
  mutate(Starch = factor(Starch,
                         levels = lvl),
         ABS=as.numeric(ABS))
kable(UV2)
```

Starch	sub_conc	Time	ABS
Soluble Starch	1.0	0	0.9021
Soluble Starch	1.0	15	0.4526
Soluble Starch	1.0	30	0.2145
Soluble Starch	1.0	45	0.0124
Soluble Starch	1.0	60	0.0102
Soluble Starch	1.0	0	0.8968
Soluble Starch	1.0	15	0.4469
Soluble Starch	1.0	30	0.2078
Soluble Starch	1.0	45	0.0111
Soluble Starch	1.0	60	0.0095
Topioca Starch	1.0	0	1.0618
Topioca Starch	1.0	15	0.6215
Topioca Starch	1.0	30	0.3574

Starch	sub_conc	Time	ABS
Topioca Starch	1.0	45	0.1401
Topioca Starch	1.0	60	0.0935
Topioca Starch	1.0	0	1.0576
Topioca Starch	1.0	15	0.6188
Topioca Starch	1.0	30	0.3559
Topioca Starch	1.0	45	0.1368
Topioca Starch	1.0	60	0.0962
Corn Starch	1.0	0	0.9842
Corn Starch	1.0	15	0.5059
Corn Starch	1.0	30	0.2764
Corn Starch	1.0	45	0.0187
Corn Starch	1.0	60	0.0139
Corn Starch	1.0	0	0.9915
Corn Starch	1.0	15	0.5126
Corn Starch	1.0	30	0.2698
Corn Starch	1.0	45	0.0202
Corn Starch	1.0	60	0.0147
Rice Starch	1.0	0	1.1250
Rice Starch	1.0	15	0.6748
Rice Starch	1.0	30	0.3629
Rice Starch	1.0	45	0.0524
Rice Starch	1.0	60	0.0426
Rice Starch	1.0	0	1.3380
Rice Starch	1.0	15	0.6785
Rice Starch	1.0	30	0.3859
Rice Starch	1.0	45	0.0479
Rice Starch	1.0	60	0.0335
Negative control	1.0	0	0.9328
Negative control	1.0	15	0.9326
Negative control	1.0	30	0.9331
Negative control	1.0	45	0.9321
Negative control	1.0	60	0.9348
Soluble Starch	0.5	0	0.5124
Soluble Starch	0.5	15	0.2866
Soluble Starch	0.5	30	0.1428
Soluble Starch	0.5	45	0.0084
Soluble Starch	0.5	60	0.0068
Soluble Starch	0.5	0	0.5181
Soluble Starch	0.5	15	0.2818
Soluble Starch	0.5	30	0.1379
Soluble Starch	0.5	45	0.0097
Soluble Starch	0.5	60	0.0059
Topioca Starch	0.5	0	0.6128

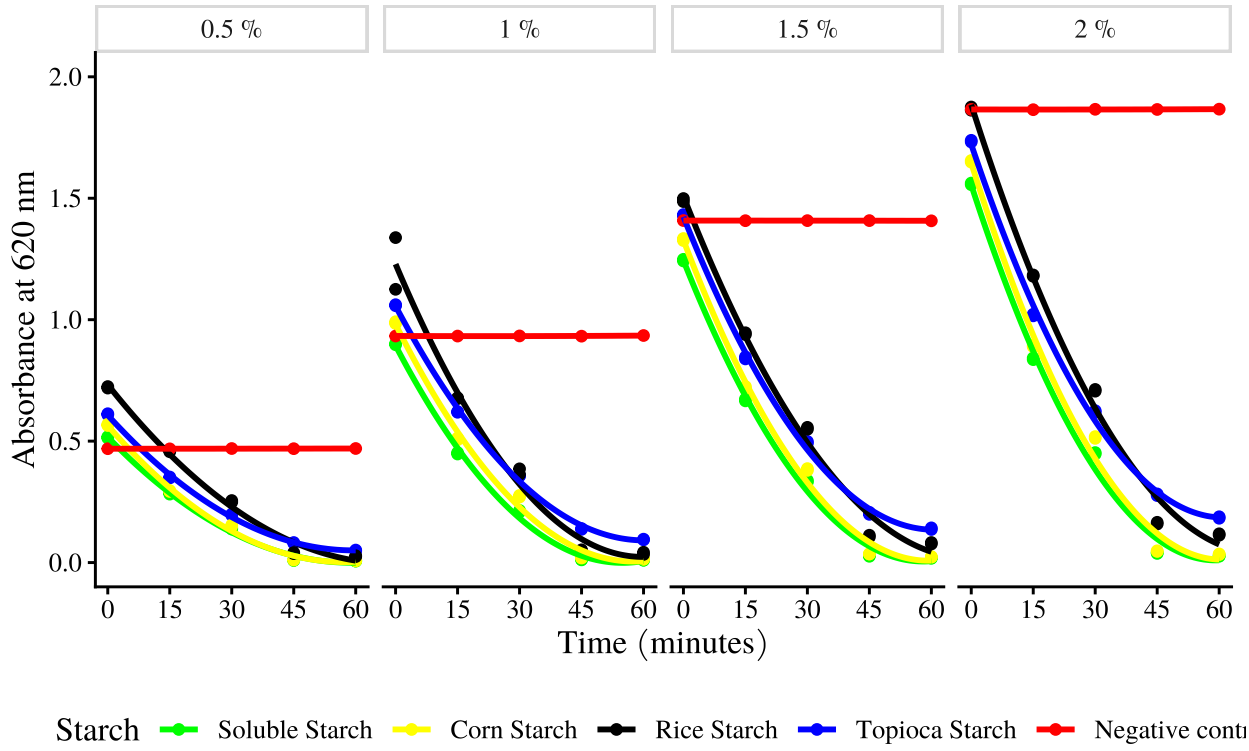
Starch	sub_conc	Time	ABS
Topioca Starch	0.5	15	0.3522
Topioca Starch	0.5	30	0.1998
Topioca Starch	0.5	45	0.0825
Topioca Starch	0.5	60	0.0514
Topioca Starch	0.5	0	0.6085
Topioca Starch	0.5	15	0.3484
Topioca Starch	0.5	30	0.1948
Topioca Starch	0.5	45	0.0796
Topioca Starch	0.5	60	0.0478
Corn Starch	0.5	0	0.5642
Corn Starch	0.5	15	0.2928
Corn Starch	0.5	30	0.1546
Corn Starch	0.5	45	0.0104
Corn Starch	0.5	60	0.0082
Corn Starch	0.5	0	0.5695
Corn Starch	0.5	15	0.2964
Corn Starch	0.5	30	0.1498
Corn Starch	0.5	45	0.0118
Corn Starch	0.5	60	0.0071
Rice Starch	0.5	0	0.7184
Rice Starch	0.5	15	0.4562
Rice Starch	0.5	30	0.2485
Rice Starch	0.5	45	0.0368
Rice Starch	0.5	60	0.0245
Rice Starch	0.5	0	0.7246
Rice Starch	0.5	15	0.4628
Rice Starch	0.5	30	0.2554
Rice Starch	0.5	45	0.0412
Rice Starch	0.5	60	0.0281
Negative control	0.5	0	0.4688
Negative control	0.5	15	0.4682
Negative control	0.5	30	0.4695
Negative control	0.5	45	0.4689
Negative control	0.5	60	0.4696
Soluble Starch	1.5	0	1.2486
Soluble Starch	1.5	15	0.6735
Soluble Starch	1.5	30	0.3387
Soluble Starch	1.5	45	0.0286
Soluble Starch	1.5	60	0.0198
Soluble Starch	1.5	0	1.2418
Soluble Starch	1.5	15	0.6658
Soluble Starch	1.5	30	0.3314
Soluble Starch	1.5	45	0.0259

Starch	sub_conc	Time	ABS
Soluble Starch	1.5	60	0.0176
Topioca Starch	1.5	0	1.4328
Topioca Starch	1.5	15	0.8468
Topioca Starch	1.5	30	0.4986
Topioca Starch	1.5	45	0.2068
Topioca Starch	1.5	60	0.1428
Topioca Starch	1.5	0	1.4268
Topioca Starch	1.5	15	0.8385
Topioca Starch	1.5	30	0.4912
Topioca Starch	1.5	45	0.1985
Topioca Starch	1.5	60	0.1362
Corn Starch	1.5	0	1.3258
Corn Starch	1.5	15	0.7165
Corn Starch	1.5	30	0.3868
Corn Starch	1.5	45	0.0352
Corn Starch	1.5	60	0.0248
Corn Starch	1.5	0	1.3342
Corn Starch	1.5	15	0.7248
Corn Starch	1.5	30	0.3796
Corn Starch	1.5	45	0.0358
Corn Starch	1.5	60	0.0221
Rice Starch	1.5	0	1.4862
Rice Starch	1.5	15	0.9386
Rice Starch	1.5	30	0.5568
Rice Starch	1.5	45	0.1126
Rice Starch	1.5	60	0.0826
Rice Starch	1.5	0	1.4988
Rice Starch	1.5	15	0.9462
Rice Starch	1.5	30	0.5486
Rice Starch	1.5	45	0.1048
Rice Starch	1.5	60	0.0765
Negative control	1.5	0	1.4082
Negative control	1.5	15	1.4076
Negative control	1.5	30	1.4078
Negative control	1.5	45	1.4080
Negative control	1.5	60	1.4070
Soluble Starch	2.0	0	1.5624
Soluble Starch	2.0	15	0.8426
Soluble Starch	2.0	30	0.4528
Soluble Starch	2.0	45	0.0412
Soluble Starch	2.0	60	0.0284
Soluble Starch	2.0	0	1.5558
Soluble Starch	2.0	15	0.8354

Starch	sub_conc	Time	ABS
Soluble Starch	2.0	30	0.4465
Soluble Starch	2.0	45	0.0376
Soluble Starch	2.0	60	0.0259
Topioca Starch	2.0	0	1.7385
Topioca Starch	2.0	15	1.0224
Topioca Starch	2.0	30	0.6254
Topioca Starch	2.0	45	0.2826
Topioca Starch	2.0	60	0.1885
Topioca Starch	2.0	0	1.7316
Topioca Starch	2.0	15	1.0158
Topioca Starch	2.0	30	0.6187
Topioca Starch	2.0	45	0.2758
Topioca Starch	2.0	60	0.1824
Corn Starch	2.0	0	1.6488
Corn Starch	2.0	15	0.8958
Corn Starch	2.0	30	0.5186
Corn Starch	2.0	45	0.0485
Corn Starch	2.0	60	0.0354
Corn Starch	2.0	0	1.6554
Corn Starch	2.0	15	0.8886
Corn Starch	2.0	30	0.5114
Corn Starch	2.0	45	0.0458
Corn Starch	2.0	60	0.0328
Rice Starch	2.0	0	1.8624
Rice Starch	2.0	15	1.1842
Rice Starch	2.0	30	0.7128
Rice Starch	2.0	45	0.1658
Rice Starch	2.0	60	0.1184
Rice Starch	2.0	0	1.8756
Rice Starch	2.0	15	1.1786
Rice Starch	2.0	30	0.7056
Rice Starch	2.0	45	0.1586
Rice Starch	2.0	60	0.1126
Negative control	2.0	0	1.8658
Negative control	2.0	15	1.8646
Negative control	2.0	30	1.8662
Negative control	2.0	45	1.8655
Negative control	2.0	60	1.8668

6.1 Visualizing the enzyme activity

```
UV2 %>%
  #filter(Starch!="Negative control")%>%
  ggplot(aes(x=Time, y=ABS, color = Starch))+
  geom_point()+
  coord_cartesian(ylim = c(0,2))+
  geom_smooth(method = 'lm', se = F,
              formula = y~poly(x, 2, raw =T))+
  scale_x_continuous(breaks = seq(0, 60, 15))+
  scale_y_continuous(breaks = seq(0, 2, 0.5))+
  scale_color_manual(
    breaks = lvl,
    values = c("green",
               "yellow",
               "black",
               "blue",
               "red")
  )+
  labs(
    x = "Time (minutes)",
    y = "Absorbance at 620 nm"
  )+
  facet_grid(.~sub_conc,
             labeller = as_labeller(function(x){
               paste(x, "%")
             }))+
  theme_classic()+
  theme(
    text = element_text(family= mfont,
                        size = 12,
                        color = 'black'),
    legend.position = "bottom",
    strip.background = element_rect(color = "grey85")
    )
```




```
UV2.1 <- UV2 %>% reframe(Absmean=mean(ABS),
                         .by = c(Time, sub_conc, Starch)) %>%
  # group_by(Starch, Time, sub_conc) %>%
  mutate(ABSper=(Absmean)/first(Absmean)*100,
         .by = c(sub_conc, Starch))
kable(UV2.1)
```

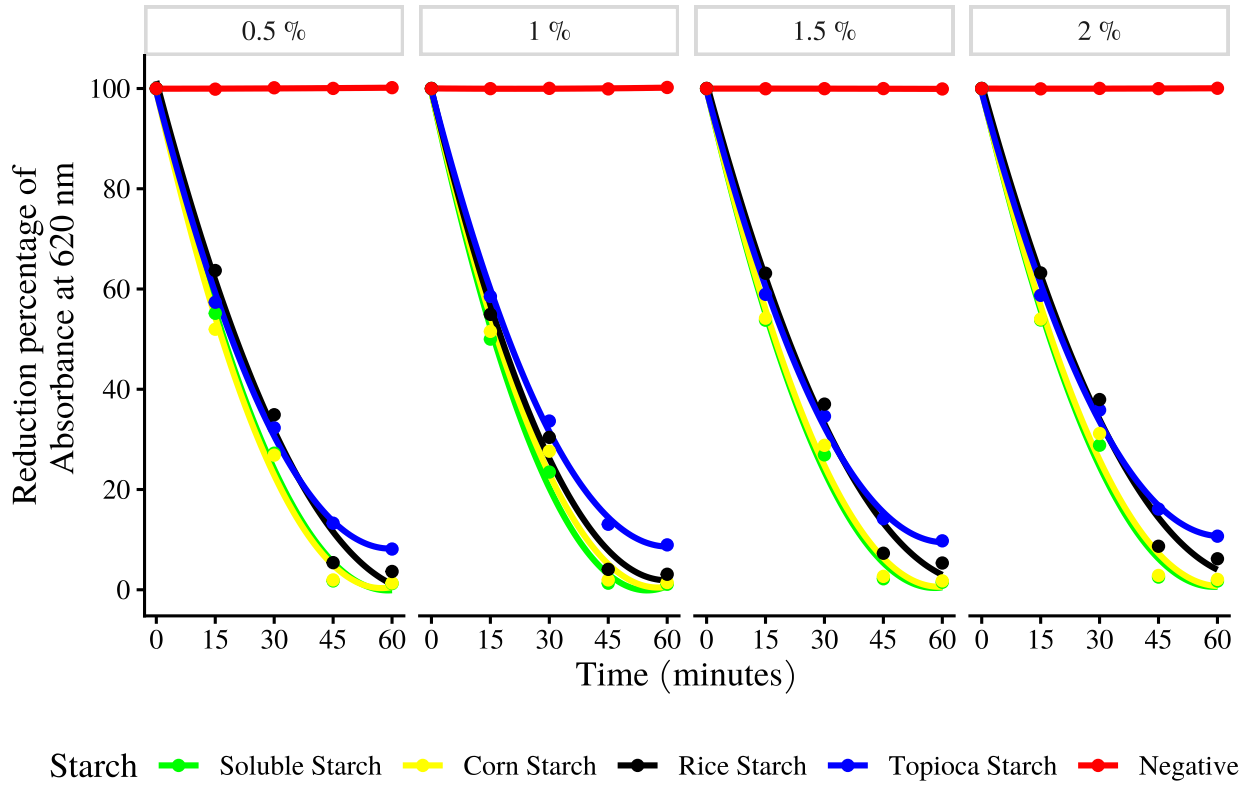
Time	sub_conc	Starch	Absmean	ABSper
0	1.0	Soluble Starch	0.89945	100.000000
15	1.0	Soluble Starch	0.44975	50.002780
30	1.0	Soluble Starch	0.21115	23.475457
45	1.0	Soluble Starch	0.01175	1.306354
60	1.0	Soluble Starch	0.00985	1.095114
0	1.0	Topioca Starch	1.05970	100.000000
15	1.0	Topioca Starch	0.62015	58.521280
30	1.0	Topioca Starch	0.35665	33.655752
45	1.0	Topioca Starch	0.13845	13.065018
60	1.0	Topioca Starch	0.09485	8.950646
0	1.0	Corn Starch	0.98785	100.000000
15	1.0	Corn Starch	0.50925	51.551349
30	1.0	Corn Starch	0.27310	27.645898

Time	sub_conc	Starch	Absmean	ABSper
45	1.0	Corn Starch	0.01945	1.968922
60	1.0	Corn Starch	0.01430	1.447588
0	1.0	Rice Starch	1.23150	100.000000
15	1.0	Rice Starch	0.67665	54.945189
30	1.0	Rice Starch	0.37440	30.401949
45	1.0	Rice Starch	0.05015	4.072270
60	1.0	Rice Starch	0.03805	3.089728
0	1.0	Negative control	0.93280	100.000000
15	1.0	Negative control	0.93260	99.978559
30	1.0	Negative control	0.93310	100.032161
45	1.0	Negative control	0.93210	99.924957
60	1.0	Negative control	0.93480	100.214408
0	0.5	Soluble Starch	0.51525	100.000000
15	0.5	Soluble Starch	0.28420	55.157690
30	0.5	Soluble Starch	0.14035	27.239204
45	0.5	Soluble Starch	0.00905	1.756429
60	0.5	Soluble Starch	0.00635	1.232411
0	0.5	Topioca Starch	0.61065	100.000000
15	0.5	Topioca Starch	0.35030	57.365103
30	0.5	Topioca Starch	0.19730	32.309834
45	0.5	Topioca Starch	0.08105	13.272742
60	0.5	Topioca Starch	0.04960	8.122492
0	0.5	Corn Starch	0.56685	100.000000
15	0.5	Corn Starch	0.29460	51.971421
30	0.5	Corn Starch	0.15220	26.850137
45	0.5	Corn Starch	0.01110	1.958190
60	0.5	Corn Starch	0.00765	1.349563
0	0.5	Rice Starch	0.72150	100.000000
15	0.5	Rice Starch	0.45950	63.686764
30	0.5	Rice Starch	0.25195	34.920305
45	0.5	Rice Starch	0.03900	5.405405
60	0.5	Rice Starch	0.02630	3.645184
0	0.5	Negative control	0.46880	100.000000
15	0.5	Negative control	0.46820	99.872014
30	0.5	Negative control	0.46950	100.149317
45	0.5	Negative control	0.46890	100.021331
60	0.5	Negative control	0.46960	100.170648
0	1.5	Soluble Starch	1.24520	100.000000
15	1.5	Soluble Starch	0.66965	53.778509
30	1.5	Soluble Starch	0.33505	26.907324
45	1.5	Soluble Starch	0.02725	2.188404
60	1.5	Soluble Starch	0.01870	1.501767
0	1.5	Topioca Starch	1.42980	100.000000

Time	sub_conc	Starch	Absmean	ABSper
15	1.5	Topioca Starch	0.84265	58.934816
30	1.5	Topioca Starch	0.49490	34.613233
45	1.5	Topioca Starch	0.20265	14.173311
60	1.5	Topioca Starch	0.13950	9.756609
0	1.5	Corn Starch	1.33000	100.000000
15	1.5	Corn Starch	0.72065	54.184210
30	1.5	Corn Starch	0.38320	28.812030
45	1.5	Corn Starch	0.03550	2.669173
60	1.5	Corn Starch	0.02345	1.763158
0	1.5	Rice Starch	1.49250	100.000000
15	1.5	Rice Starch	0.94240	63.142379
30	1.5	Rice Starch	0.55270	37.031826
45	1.5	Rice Starch	0.10870	7.283082
60	1.5	Rice Starch	0.07955	5.329983
0	1.5	Negative control	1.40820	100.000000
15	1.5	Negative control	1.40760	99.957392
30	1.5	Negative control	1.40780	99.971595
45	1.5	Negative control	1.40800	99.985798
60	1.5	Negative control	1.40700	99.914785
0	2.0	Soluble Starch	1.55910	100.000000
15	2.0	Soluble Starch	0.83900	53.813097
30	2.0	Soluble Starch	0.44965	28.840357
45	2.0	Soluble Starch	0.03940	2.527099
60	2.0	Soluble Starch	0.02715	1.741389
0	2.0	Topioca Starch	1.73505	100.000000
15	2.0	Topioca Starch	1.01910	58.736060
30	2.0	Topioca Starch	0.62205	35.851993
45	2.0	Topioca Starch	0.27920	16.091755
60	2.0	Topioca Starch	0.18545	10.688453
0	2.0	Corn Starch	1.65210	100.000000
15	2.0	Corn Starch	0.89220	54.003995
30	2.0	Corn Starch	0.51500	31.172447
45	2.0	Corn Starch	0.04715	2.853944
60	2.0	Corn Starch	0.03410	2.064040
0	2.0	Rice Starch	1.86900	100.000000
15	2.0	Rice Starch	1.18140	63.210273
30	2.0	Rice Starch	0.70920	37.945425
45	2.0	Rice Starch	0.16220	8.678438
60	2.0	Rice Starch	0.11550	6.179775
0	2.0	Negative control	1.86580	100.000000
15	2.0	Negative control	1.86460	99.935684
30	2.0	Negative control	1.86620	100.021439
45	2.0	Negative control	1.86550	99.983921

Time	sub_conc	Starch	Absmean	ABSper
60	2.0	Negative control	1.86680	100.053596

```
UV2.1%>%
  ggplot(aes(x=Time, y=ABSper, color = Starch))+
  #geom_smooth()
  geom_smooth(method = 'lm', se = F,
              formula = y~poly(x, 2, raw =T))+
  geom_point()+
  scale_x_continuous(breaks = seq(0, 60, 15))+
  scale_y_continuous(breaks = seq(0, 100, 20))+
  scale_color_manual(
    breaks = lvl,
    values = c("green",
               "yellow",
               "black",
               "blue",
               "red")
  )+
  labs(
    x = "Time (minutes)",
    y = "Reduction percentage of
    Absorbance at 620 nm"
  )+
  facet_grid(.~sub_conc,
             labeller = as_labeller(function(x){
               paste(x, "%")
             }))+
  theme_classic()+
  theme(
    text = element_text(family= mfont,
                        size = 12,
                        color = 'black'),
    legend.position = "bottom",
    strip.background = element_rect(color = "grey85")
    # ,
    # panel.grid.major = element_line(
    #   color = "grey80",
    #   linewidth = 0.3,
    #   linetype = 3
  )
```



7 Numerical estimation of the curvature of the enzyme activity

```
mod <- lm(ABS~Starch+poly(Time, 2, raw =T), data= UV2 %>%
            filter(Starch!="Negative control") )
summary(mod)
```

```
Call:
lm(formula = ABS ~ Starch + poly(Time, 2, raw = T), data = UV2 %>%
    filter(Starch != "Negative control"))

Residuals:
    Min      1Q  Median      3Q     Max
-0.6249 -0.1074 -0.0081  0.1042  0.6136

Coefficients:
                          Estimate Std. Error t value Pr(>|t|)
(Intercept)              1.095e+00  5.011e-02  21.859  < 2e-16 ***
StarchCorn Starch        3.612e-02  5.200e-02   0.695  0.48832
StarchRice Starch        1.667e-01  5.200e-02   3.207  0.00163 **
```

```
StarchTopioca Starch     1.381e-01  5.200e-02   2.656  0.00875 **
poly(Time, 2, raw = T)1 -3.744e-02  3.055e-03 -12.253  < 2e-16 ***
poly(Time, 2, raw = T)2  3.076e-04  4.883e-05   6.299    3e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2325 on 154 degrees of freedom
Multiple R-squared:  0.776, Adjusted R-squared:  0.7688
F-statistic: 106.7 on 5 and 154 DF,  p-value: < 2.2e-16
```

```r
# estimattion vs lab-tested value
uv3 <- UV2 %>% filter(Starch != "Negative control")
uv3$fitted <- fitted(mod)
uv3$residual <- resid(mod)
kable(uv3)
```

Starch	sub_conc	Time	ABS	fitted	residual
Soluble Starch	1.0	0	0.9021	1.0952586	-0.1931586
Soluble Starch	1.0	15	0.4526	0.6029145	-0.1503145
Soluble Starch	1.0	30	0.2145	0.2489677	-0.0344677
Soluble Starch	1.0	45	0.0124	0.0334182	-0.0210182
Soluble Starch	1.0	60	0.0102	-0.0437339	0.0539339
Soluble Starch	1.0	0	0.8968	1.0952586	-0.1984586
Soluble Starch	1.0	15	0.4469	0.6029145	-0.1560145
Soluble Starch	1.0	30	0.2078	0.2489677	-0.0411677
Soluble Starch	1.0	45	0.0111	0.0334182	-0.0223182
Soluble Starch	1.0	60	0.0095	-0.0437339	0.0532339
Topioca Starch	1.0	0	1.0618	1.2333461	-0.1715461
Topioca Starch	1.0	15	0.6215	0.7410020	-0.1195020
Topioca Starch	1.0	30	0.3574	0.3870552	-0.0296552
Topioca Starch	1.0	45	0.1401	0.1715057	-0.0314057
Topioca Starch	1.0	60	0.0935	0.0943536	-0.0008536
Topioca Starch	1.0	0	1.0576	1.2333461	-0.1757461
Topioca Starch	1.0	15	0.6188	0.7410020	-0.1222020
Topioca Starch	1.0	30	0.3559	0.3870552	-0.0311552
Topioca Starch	1.0	45	0.1368	0.1715057	-0.0347057
Topioca Starch	1.0	60	0.0962	0.0943536	0.0018464
Corn Starch	1.0	0	0.9842	1.1313786	-0.1471786
Corn Starch	1.0	15	0.5059	0.6390345	-0.1331345
Corn Starch	1.0	30	0.2764	0.2850877	-0.0086877
Corn Starch	1.0	45	0.0187	0.0695382	-0.0508382
Corn Starch	1.0	60	0.0139	-0.0076139	0.0215139
Corn Starch	1.0	0	0.9915	1.1313786	-0.1398786

Starch	sub_conc	Time	ABS	fitted	residual
Corn Starch	1.0	15	0.5126	0.6390345	-0.1264345
Corn Starch	1.0	30	0.2698	0.2850877	-0.0152877
Corn Starch	1.0	45	0.0202	0.0695382	-0.0493382
Corn Starch	1.0	60	0.0147	-0.0076139	0.0223139
Rice Starch	1.0	0	1.1250	1.2620011	-0.1370011
Rice Starch	1.0	15	0.6748	0.7696570	-0.0948570
Rice Starch	1.0	30	0.3629	0.4157102	-0.0528102
Rice Starch	1.0	45	0.0524	0.2001607	-0.1477607
Rice Starch	1.0	60	0.0426	0.1230086	-0.0804086
Rice Starch	1.0	0	1.3380	1.2620011	0.0759989
Rice Starch	1.0	15	0.6785	0.7696570	-0.0911570
Rice Starch	1.0	30	0.3859	0.4157102	-0.0298102
Rice Starch	1.0	45	0.0479	0.2001607	-0.1522607
Rice Starch	1.0	60	0.0335	0.1230086	-0.0895086
Soluble Starch	0.5	0	0.5124	1.0952586	-0.5828586
Soluble Starch	0.5	15	0.2866	0.6029145	-0.3163145
Soluble Starch	0.5	30	0.1428	0.2489677	-0.1061677
Soluble Starch	0.5	45	0.0084	0.0334182	-0.0250182
Soluble Starch	0.5	60	0.0068	-0.0437339	0.0505339
Soluble Starch	0.5	0	0.5181	1.0952586	-0.5771586
Soluble Starch	0.5	15	0.2818	0.6029145	-0.3211145
Soluble Starch	0.5	30	0.1379	0.2489677	-0.1110677
Soluble Starch	0.5	45	0.0097	0.0334182	-0.0237182
Soluble Starch	0.5	60	0.0059	-0.0437339	0.0496339
Topioca Starch	0.5	0	0.6128	1.2333461	-0.6205461
Topioca Starch	0.5	15	0.3522	0.7410020	-0.3888020
Topioca Starch	0.5	30	0.1998	0.3870552	-0.1872552
Topioca Starch	0.5	45	0.0825	0.1715057	-0.0890057
Topioca Starch	0.5	60	0.0514	0.0943536	-0.0429536
Topioca Starch	0.5	0	0.6085	1.2333461	-0.6248461
Topioca Starch	0.5	15	0.3484	0.7410020	-0.3926020
Topioca Starch	0.5	30	0.1948	0.3870552	-0.1922552
Topioca Starch	0.5	45	0.0796	0.1715057	-0.0919057
Topioca Starch	0.5	60	0.0478	0.0943536	-0.0465536
Corn Starch	0.5	0	0.5642	1.1313786	-0.5671786
Corn Starch	0.5	15	0.2928	0.6390345	-0.3462345
Corn Starch	0.5	30	0.1546	0.2850877	-0.1304877
Corn Starch	0.5	45	0.0104	0.0695382	-0.0591382
Corn Starch	0.5	60	0.0082	-0.0076139	0.0158139
Corn Starch	0.5	0	0.5695	1.1313786	-0.5618786
Corn Starch	0.5	15	0.2964	0.6390345	-0.3426345
Corn Starch	0.5	30	0.1498	0.2850877	-0.1352877
Corn Starch	0.5	45	0.0118	0.0695382	-0.0577382

Starch	sub_conc	Time	ABS	fitted	residual
Corn Starch	0.5	60	0.0071	-0.0076139	0.0147139
Rice Starch	0.5	0	0.7184	1.2620011	-0.5436011
Rice Starch	0.5	15	0.4562	0.7696570	-0.3134570
Rice Starch	0.5	30	0.2485	0.4157102	-0.1672102
Rice Starch	0.5	45	0.0368	0.2001607	-0.1633607
Rice Starch	0.5	60	0.0245	0.1230086	-0.0985086
Rice Starch	0.5	0	0.7246	1.2620011	-0.5374011
Rice Starch	0.5	15	0.4628	0.7696570	-0.3068570
Rice Starch	0.5	30	0.2554	0.4157102	-0.1603102
Rice Starch	0.5	45	0.0412	0.2001607	-0.1589607
Rice Starch	0.5	60	0.0281	0.1230086	-0.0949086
Soluble Starch	1.5	0	1.2486	1.0952586	0.1533414
Soluble Starch	1.5	15	0.6735	0.6029145	0.0705855
Soluble Starch	1.5	30	0.3387	0.2489677	0.0897323
Soluble Starch	1.5	45	0.0286	0.0334182	-0.0048182
Soluble Starch	1.5	60	0.0198	-0.0437339	0.0635339
Soluble Starch	1.5	0	1.2418	1.0952586	0.1465414
Soluble Starch	1.5	15	0.6658	0.6029145	0.0628855
Soluble Starch	1.5	30	0.3314	0.2489677	0.0824323
Soluble Starch	1.5	45	0.0259	0.0334182	-0.0075182
Soluble Starch	1.5	60	0.0176	-0.0437339	0.0613339
Topioca Starch	1.5	0	1.4328	1.2333461	0.1994539
Topioca Starch	1.5	15	0.8468	0.7410020	0.1057980
Topioca Starch	1.5	30	0.4986	0.3870552	0.1115448
Topioca Starch	1.5	45	0.2068	0.1715057	0.0352943
Topioca Starch	1.5	60	0.1428	0.0943536	0.0484464
Topioca Starch	1.5	0	1.4268	1.2333461	0.1934539
Topioca Starch	1.5	15	0.8385	0.7410020	0.0974980
Topioca Starch	1.5	30	0.4912	0.3870552	0.1041448
Topioca Starch	1.5	45	0.1985	0.1715057	0.0269943
Topioca Starch	1.5	60	0.1362	0.0943536	0.0418464
Corn Starch	1.5	0	1.3258	1.1313786	0.1944214
Corn Starch	1.5	15	0.7165	0.6390345	0.0774655
Corn Starch	1.5	30	0.3868	0.2850877	0.1017123
Corn Starch	1.5	45	0.0352	0.0695382	-0.0343382
Corn Starch	1.5	60	0.0248	-0.0076139	0.0324139
Corn Starch	1.5	0	1.3342	1.1313786	0.2028214
Corn Starch	1.5	15	0.7248	0.6390345	0.0857655
Corn Starch	1.5	30	0.3796	0.2850877	0.0945123
Corn Starch	1.5	45	0.0358	0.0695382	-0.0337382
Corn Starch	1.5	60	0.0221	-0.0076139	0.0297139
Rice Starch	1.5	0	1.4862	1.2620011	0.2241989
Rice Starch	1.5	15	0.9386	0.7696570	0.1689430

Starch	sub_conc	Time	ABS	fitted	residual
Rice Starch	1.5	30	0.5568	0.4157102	0.1410898
Rice Starch	1.5	45	0.1126	0.2001607	-0.0875607
Rice Starch	1.5	60	0.0826	0.1230086	-0.0404086
Rice Starch	1.5	0	1.4988	1.2620011	0.2367989
Rice Starch	1.5	15	0.9462	0.7696570	0.1765430
Rice Starch	1.5	30	0.5486	0.4157102	0.1328898
Rice Starch	1.5	45	0.1048	0.2001607	-0.0953607
Rice Starch	1.5	60	0.0765	0.1230086	-0.0465086
Soluble Starch	2.0	0	1.5624	1.0952586	0.4671414
Soluble Starch	2.0	15	0.8426	0.6029145	0.2396855
Soluble Starch	2.0	30	0.4528	0.2489677	0.2038323
Soluble Starch	2.0	45	0.0412	0.0334182	0.0077818
Soluble Starch	2.0	60	0.0284	-0.0437339	0.0721339
Soluble Starch	2.0	0	1.5558	1.0952586	0.4605414
Soluble Starch	2.0	15	0.8354	0.6029145	0.2324855
Soluble Starch	2.0	30	0.4465	0.2489677	0.1975323
Soluble Starch	2.0	45	0.0376	0.0334182	0.0041818
Soluble Starch	2.0	60	0.0259	-0.0437339	0.0696339
Topioca Starch	2.0	0	1.7385	1.2333461	0.5051539
Topioca Starch	2.0	15	1.0224	0.7410020	0.2813980
Topioca Starch	2.0	30	0.6254	0.3870552	0.2383448
Topioca Starch	2.0	45	0.2826	0.1715057	0.1110943
Topioca Starch	2.0	60	0.1885	0.0943536	0.0941464
Topioca Starch	2.0	0	1.7316	1.2333461	0.4982539
Topioca Starch	2.0	15	1.0158	0.7410020	0.2747980
Topioca Starch	2.0	30	0.6187	0.3870552	0.2316448
Topioca Starch	2.0	45	0.2758	0.1715057	0.1042943
Topioca Starch	2.0	60	0.1824	0.0943536	0.0880464
Corn Starch	2.0	0	1.6488	1.1313786	0.5174214
Corn Starch	2.0	15	0.8958	0.6390345	0.2567655
Corn Starch	2.0	30	0.5186	0.2850877	0.2335123
Corn Starch	2.0	45	0.0485	0.0695382	-0.0210382
Corn Starch	2.0	60	0.0354	-0.0076139	0.0430139
Corn Starch	2.0	0	1.6554	1.1313786	0.5240214
Corn Starch	2.0	15	0.8886	0.6390345	0.2495655
Corn Starch	2.0	30	0.5114	0.2850877	0.2263123
Corn Starch	2.0	45	0.0458	0.0695382	-0.0237382
Corn Starch	2.0	60	0.0328	-0.0076139	0.0404139
Rice Starch	2.0	0	1.8624	1.2620011	0.6003989
Rice Starch	2.0	15	1.1842	0.7696570	0.4145430
Rice Starch	2.0	30	0.7128	0.4157102	0.2970898
Rice Starch	2.0	45	0.1658	0.2001607	-0.0343607
Rice Starch	2.0	60	0.1184	0.1230086	-0.0046086

Starch	sub_conc	Time	ABS	fitted	residual
Rice Starch	2.0	0	1.8756	1.2620011	0.6135989
Rice Starch	2.0	15	1.1786	0.7696570	0.4089430
Rice Starch	2.0	30	0.7056	0.4157102	0.2898898
Rice Starch	2.0	45	0.1586	0.2001607	-0.0415607
Rice Starch	2.0	60	0.1126	0.1230086	-0.0104086

8 Estimation of kinetic parameters Vo, Km and Vmax

```
# 1. Calculate Initial Velocity (v0)
# We calculate v0 using the change btw Time = 0 and Time = 15
v0_data <- UV2 %>%
  # Exclude the negative control
  filter(Starch != "Negative control") %>%
  # Filter only for the initial linear phase timepoints
  filter(Time %in% c(0, 15)) %>%
  # Group to average the two replicates in your dataset
  group_by(Starch, sub_conc, Time) %>%
  summarize(mean_ABS = mean(ABS), .groups = "drop") %>%
  # Reshape data to have Time 0 and Time 15 in separate columns
  pivot_wider(names_from = Time, values_from = mean_ABS,
              names_prefix = "T_") %>%
  mutate(
    # Calculate initial velocity
    v0 = (T_0 - T_15) / 15
  )

print("Calculated Initial Velocities (v0):")
```

```
[1] "Calculated Initial Velocities (v0):"
```

```
kable(v0_data, format = 'latex', booktabs = TRUE, escape = T)
```

Starch	sub_conc	T_0	T_15	v0
Soluble Starch	0.5	0.51525	0.28420	0.0154033
Soluble Starch	1.0	0.89945	0.44975	0.0299800
Soluble Starch	1.5	1.24520	0.66965	0.0383700
Soluble Starch	2.0	1.55910	0.83900	0.0480067
Corn Starch	0.5	0.56685	0.29460	0.0181500
Corn Starch	1.0	0.98785	0.50925	0.0319067
Corn Starch	1.5	1.33000	0.72065	0.0406233
Corn Starch	2.0	1.65210	0.89220	0.0506600
Rice Starch	0.5	0.72150	0.45950	0.0174667
Rice Starch	1.0	1.23150	0.67665	0.0369900
Rice Starch	1.5	1.49250	0.94240	0.0366733
Rice Starch	2.0	1.86900	1.18140	0.0458400
Topioca Starch	0.5	0.61065	0.35030	0.0173567
Topioca Starch	1.0	1.05970	0.62015	0.0293033
Topioca Starch	1.5	1.42980	0.84265	0.0391433
Topioca Starch	2.0	1.73505	1.01910	0.0477300

8.1 Fitting Michaeli-Menten Model for the four types of starch

```
mm_results <- v0_data %>%
  group_by(Starch) %>%
  nest() %>%
  mutate(
    model = map(data, ~ nls(
      v0 ~ (Vmax * sub_conc) / (Km + sub_conc),
      data = .x,
      # Dynamic start values based on the data
      start = list(Vmax = max(.x$v0), Km = median(.x$sub_conc))
    )),
    # Extract the Vmax and Km estimates cleanly
    params = map(model, tidy)
  ) %>%
  unnest(params) %>%
  select(Starch, term, estimate, std.error, p.value)

print("Estimated Vmax and Km parameters:")
```

```
[1] "Estimated Vmax and Km parameters:"
```

```
kable(mm_results)
```

Starch	term	estimate	std.error	p.value
Soluble Starch	Vmax	0.1394114	0.0248543	0.0303446
Soluble Starch	Km	3.8346015	0.9569834	0.0570089
Corn Starch	Vmax	0.1237107	0.0148865	0.0141730
Corn Starch	Km	2.9409360	0.5344320	0.0314722
Rice Starch	Vmax	0.0772405	0.0263232	0.0991661
Rice Starch	Km	1.4089156	0.9497896	0.2762165
Topioca Starch	Vmax	0.1199012	0.0080030	0.0044255
Topioca Starch	Km	3.0542478	0.3043381	0.0097835

```r
## Another way of fitting MM model
suppressMessages(
  suppressWarnings(
    library(drc, quietly = TRUE)
  )
)
fit <- drm(v0~sub_conc, data = v0_data, fct = MM.2())
summary(fit)
```

```
Model fitted: Michaelis-Menten (2 parms)

Parameter estimates:

              Estimate Std. Error t-value   p-value
d:(Intercept) 0.107786   0.014422  7.4736 2.996e-06 ***
e:(Intercept) 2.530430   0.536467  4.7168 0.0003305 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error:

 0.002395029 (14 degrees of freedom)
```

```r
# Fit separate MM curves for each Starch group
fit_grouped <- drm(v0 ~ sub_conc, curveid = Starch,
                   data = v0_data, fct = MM.2())

# Summary of Vmax (d) and Km (e) for each starch type
summary(fit_grouped)
```

```
Model fitted: Michaelis-Menten (2 parms)
```

```
Parameter estimates:

                 Estimate Std. Error t-value   p-value
d:Soluble Starch 0.138942   0.053124  2.6154 0.0308689 *
d:Corn Starch    0.123652   0.036633  3.3754 0.0097078 **
d:Rice Starch    0.077240   0.013407  5.7611 0.0004236 ***
d:Topioca Starch 0.119825   0.038667  3.0989 0.0146907 *
e:Soluble Starch 3.816484   2.045083  1.8662 0.0989871 .
e:Corn Starch    2.938814   1.315300  2.2343 0.0559180 .
e:Rice Starch    1.408915   0.483061  2.9166 0.0193907 *
e:Topioca Starch 3.051361   1.470592  2.0749 0.0716789 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error:

 0.002427663 (8 degrees of freedom)
```

```r
# Predict smooth curves from fit_grouped model
pred_grid <- expand.grid(
  sub_conc = seq(0.1, 2.5, length.out = 100),
  Starch = unique(v0_data$Starch)
)
pred_grid$pred_v0 <- predict(fit_grouped, newdata = pred_grid)
kable(pred_grid, type = 'latex', escape = T, booktabs= TRUE)
```

sub_conc	Starch	pred_v0
0.1000000	Soluble Starch	0.0035476
0.1242424	Soluble Starch	0.0043805
0.1484848	Soluble Starch	0.0052033
0.1727273	Soluble Starch	0.0060160
0.1969697	Soluble Starch	0.0068189
0.2212121	Soluble Starch	0.0076122
0.2454545	Soluble Starch	0.0083960
0.2696970	Soluble Starch	0.0091705
0.2939394	Soluble Starch	0.0099358
0.3181818	Soluble Starch	0.0106922
0.3424242	Soluble Starch	0.0114398
0.3666667	Soluble Starch	0.0121787
0.3909091	Soluble Starch	0.0129091
0.4151515	Soluble Starch	0.0136311
0.4393939	Soluble Starch	0.0143449

sub_conc	Starch	pred_v0
0.4636364	Soluble Starch	0.0150506
0.4878788	Soluble Starch	0.0157484
0.5121212	Soluble Starch	0.0164383
0.5363636	Soluble Starch	0.0171206
0.5606061	Soluble Starch	0.0177953
0.5848485	Soluble Starch	0.0184625
0.6090909	Soluble Starch	0.0191225
0.6333333	Soluble Starch	0.0197753
0.6575758	Soluble Starch	0.0204210
0.6818182	Soluble Starch	0.0210597
0.7060606	Soluble Starch	0.0216916
0.7303030	Soluble Starch	0.0223167
0.7545455	Soluble Starch	0.0229353
0.7787879	Soluble Starch	0.0235473
0.8030303	Soluble Starch	0.0241528
0.8272727	Soluble Starch	0.0247521
0.8515152	Soluble Starch	0.0253451
0.8757576	Soluble Starch	0.0259320
0.9000000	Soluble Starch	0.0265129
0.9242424	Soluble Starch	0.0270878
0.9484848	Soluble Starch	0.0276568
0.9727273	Soluble Starch	0.0282202
0.9969697	Soluble Starch	0.0287778
1.0212121	Soluble Starch	0.0293298
1.0454545	Soluble Starch	0.0298764
1.0696970	Soluble Starch	0.0304175
1.0939394	Soluble Starch	0.0309533
1.1181818	Soluble Starch	0.0314838
1.1424242	Soluble Starch	0.0320091
1.1666667	Soluble Starch	0.0325293
1.1909091	Soluble Starch	0.0330445
1.2151515	Soluble Starch	0.0335547
1.2393939	Soluble Starch	0.0340600
1.2636364	Soluble Starch	0.0345605
1.2878788	Soluble Starch	0.0350563
1.3121212	Soluble Starch	0.0355473
1.3363636	Soluble Starch	0.0360338
1.3606061	Soluble Starch	0.0365156
1.3848485	Soluble Starch	0.0369930
1.4090909	Soluble Starch	0.0374660
1.4333333	Soluble Starch	0.0379346
1.4575758	Soluble Starch	0.0383989
1.4818182	Soluble Starch	0.0388589

sub_conc	Starch	pred_v0
1.5060606	Soluble Starch	0.0393147
1.5303030	Soluble Starch	0.0397665
1.5545455	Soluble Starch	0.0402141
1.5787879	Soluble Starch	0.0406577
1.6030303	Soluble Starch	0.0410973
1.6272727	Soluble Starch	0.0415331
1.6515152	Soluble Starch	0.0419649
1.6757576	Soluble Starch	0.0423930
1.7000000	Soluble Starch	0.0428173
1.7242424	Soluble Starch	0.0432378
1.7484848	Soluble Starch	0.0436547
1.7727273	Soluble Starch	0.0440680
1.7969697	Soluble Starch	0.0444778
1.8212121	Soluble Starch	0.0448840
1.8454545	Soluble Starch	0.0452867
1.8696970	Soluble Starch	0.0456860
1.8939394	Soluble Starch	0.0460819
1.9181818	Soluble Starch	0.0464744
1.9424242	Soluble Starch	0.0468637
1.9666667	Soluble Starch	0.0472496
1.9909091	Soluble Starch	0.0476324
2.0151515	Soluble Starch	0.0480120
2.0393939	Soluble Starch	0.0483884
2.0636364	Soluble Starch	0.0487617
2.0878788	Soluble Starch	0.0491320
2.1121212	Soluble Starch	0.0494993
2.1363636	Soluble Starch	0.0498635
2.1606061	Soluble Starch	0.0502248
2.1848485	Soluble Starch	0.0505832
2.2090909	Soluble Starch	0.0509386
2.2333333	Soluble Starch	0.0512913
2.2575758	Soluble Starch	0.0516411
2.2818182	Soluble Starch	0.0519882
2.3060606	Soluble Starch	0.0523325
2.3303030	Soluble Starch	0.0526740
2.3545455	Soluble Starch	0.0530129
2.3787879	Soluble Starch	0.0533492
2.4030303	Soluble Starch	0.0536828
2.4272727	Soluble Starch	0.0540138
2.4515152	Soluble Starch	0.0543423
2.4757576	Soluble Starch	0.0546682
2.5000000	Soluble Starch	0.0549917
0.1000000	Corn Starch	0.0040691

sub_conc	Starch	pred_v0
0.1242424	Corn Starch	0.0050155
0.1484848	Corn Starch	0.0059471
0.1727273	Corn Starch	0.0068641
0.1969697	Corn Starch	0.0077670
0.2212121	Corn Starch	0.0086560
0.2454545	Corn Starch	0.0095315
0.2696970	Corn Starch	0.0103938
0.2939394	Corn Starch	0.0112431
0.3181818	Corn Starch	0.0120798
0.3424242	Corn Starch	0.0129041
0.3666667	Corn Starch	0.0137163
0.3909091	Corn Starch	0.0145167
0.4151515	Corn Starch	0.0153055
0.4393939	Corn Starch	0.0160830
0.4636364	Corn Starch	0.0168495
0.4878788	Corn Starch	0.0176051
0.5121212	Corn Starch	0.0183500
0.5363636	Corn Starch	0.0190846
0.5606061	Corn Starch	0.0198090
0.5848485	Corn Starch	0.0205234
0.6090909	Corn Starch	0.0212281
0.6333333	Corn Starch	0.0219232
0.6575758	Corn Starch	0.0226089
0.6818182	Corn Starch	0.0232855
0.7060606	Corn Starch	0.0239530
0.7303030	Corn Starch	0.0246117
0.7545455	Corn Starch	0.0252618
0.7787879	Corn Starch	0.0259034
0.8030303	Corn Starch	0.0265367
0.8272727	Corn Starch	0.0271618
0.8515152	Corn Starch	0.0277790
0.8757576	Corn Starch	0.0283883
0.9000000	Corn Starch	0.0289899
0.9242424	Corn Starch	0.0295839
0.9484848	Corn Starch	0.0301705
0.9727273	Corn Starch	0.0307499
0.9969697	Corn Starch	0.0313221
1.0212121	Corn Starch	0.0318874
1.0454545	Corn Starch	0.0324457
1.0696970	Corn Starch	0.0329973
1.0939394	Corn Starch	0.0335422
1.1181818	Corn Starch	0.0340807
1.1424242	Corn Starch	0.0346127

sub_conc	Starch	pred_v0
1.1666667	Corn Starch	0.0351385
1.1909091	Corn Starch	0.0356581
1.2151515	Corn Starch	0.0361716
1.2393939	Corn Starch	0.0366792
1.2636364	Corn Starch	0.0371809
1.2878788	Corn Starch	0.0376769
1.3121212	Corn Starch	0.0381672
1.3363636	Corn Starch	0.0386519
1.3606061	Corn Starch	0.0391312
1.3848485	Corn Starch	0.0396051
1.4090909	Corn Starch	0.0400737
1.4333333	Corn Starch	0.0405371
1.4575758	Corn Starch	0.0409954
1.4818182	Corn Starch	0.0414487
1.5060606	Corn Starch	0.0418971
1.5303030	Corn Starch	0.0423405
1.5545455	Corn Starch	0.0427792
1.5787879	Corn Starch	0.0432132
1.6030303	Corn Starch	0.0436425
1.6272727	Corn Starch	0.0440673
1.6515152	Corn Starch	0.0444876
1.6757576	Corn Starch	0.0449035
1.7000000	Corn Starch	0.0453151
1.7242424	Corn Starch	0.0457223
1.7484848	Corn Starch	0.0461254
1.7727273	Corn Starch	0.0465243
1.7969697	Corn Starch	0.0469191
1.8212121	Corn Starch	0.0473099
1.8454545	Corn Starch	0.0476967
1.8696970	Corn Starch	0.0480796
1.8939394	Corn Starch	0.0484587
1.9181818	Corn Starch	0.0488340
1.9424242	Corn Starch	0.0492056
1.9666667	Corn Starch	0.0495735
1.9909091	Corn Starch	0.0499378
2.0151515	Corn Starch	0.0502985
2.0393939	Corn Starch	0.0506557
2.0636364	Corn Starch	0.0510095
2.0878788	Corn Starch	0.0513598
2.1121212	Corn Starch	0.0517068
2.1363636	Corn Starch	0.0520504
2.1606061	Corn Starch	0.0523908
2.1848485	Corn Starch	0.0527280

sub_conc	Starch	pred_v0
2.2090909	Corn Starch	0.0530620
2.2333333	Corn Starch	0.0533929
2.2575758	Corn Starch	0.0537206
2.2818182	Corn Starch	0.0540454
2.3060606	Corn Starch	0.0543671
2.3303030	Corn Starch	0.0546859
2.3545455	Corn Starch	0.0550017
2.3787879	Corn Starch	0.0553147
2.4030303	Corn Starch	0.0556248
2.4272727	Corn Starch	0.0559321
2.4515152	Corn Starch	0.0562367
2.4757576	Corn Starch	0.0565385
2.5000000	Corn Starch	0.0568377
0.1000000	Rice Starch	0.0051189
0.1242424	Rice Starch	0.0062593
0.1484848	Rice Starch	0.0073642
0.1727273	Rice Starch	0.0084352
0.1969697	Rice Starch	0.0094739
0.2212121	Rice Starch	0.0104816
0.2454545	Rice Starch	0.0114599
0.2696970	Rice Starch	0.0124099
0.2939394	Rice Starch	0.0133328
0.3181818	Rice Starch	0.0142299
0.3424242	Rice Starch	0.0151021
0.3666667	Rice Starch	0.0159504
0.3909091	Rice Starch	0.0167760
0.4151515	Rice Starch	0.0175796
0.4393939	Rice Starch	0.0183621
0.4636364	Rice Starch	0.0191243
0.4878788	Rice Starch	0.0198671
0.5121212	Rice Starch	0.0205911
0.5363636	Rice Starch	0.0212970
0.5606061	Rice Starch	0.0219856
0.5848485	Rice Starch	0.0226575
0.6090909	Rice Starch	0.0233132
0.6333333	Rice Starch	0.0239533
0.6575758	Rice Starch	0.0245784
0.6818182	Rice Starch	0.0251891
0.7060606	Rice Starch	0.0257857
0.7303030	Rice Starch	0.0263688
0.7545455	Rice Starch	0.0269388
0.7787879	Rice Starch	0.0274962
0.8030303	Rice Starch	0.0280414

sub_conc	Starch	pred_v0
0.8272727	Rice Starch	0.0285747
0.8515152	Rice Starch	0.0290967
0.8757576	Rice Starch	0.0296075
0.9000000	Rice Starch	0.0301076
0.9242424	Rice Starch	0.0305973
0.9484848	Rice Starch	0.0310770
0.9727273	Rice Starch	0.0315469
0.9969697	Rice Starch	0.0320073
1.0212121	Rice Starch	0.0324585
1.0454545	Rice Starch	0.0329009
1.0696970	Rice Starch	0.0333345
1.0939394	Rice Starch	0.0337598
1.1181818	Rice Starch	0.0341769
1.1424242	Rice Starch	0.0345861
1.1666667	Rice Starch	0.0349875
1.1909091	Rice Starch	0.0353815
1.2151515	Rice Starch	0.0357682
1.2393939	Rice Starch	0.0361479
1.2636364	Rice Starch	0.0365206
1.2878788	Rice Starch	0.0368866
1.3121212	Rice Starch	0.0372462
1.3363636	Rice Starch	0.0375993
1.3606061	Rice Starch	0.0379463
1.3848485	Rice Starch	0.0382873
1.4090909	Rice Starch	0.0386224
1.4333333	Rice Starch	0.0389518
1.4575758	Rice Starch	0.0392756
1.4818182	Rice Starch	0.0395940
1.5060606	Rice Starch	0.0399070
1.5303030	Rice Starch	0.0402150
1.5545455	Rice Starch	0.0405178
1.5787879	Rice Starch	0.0408158
1.6030303	Rice Starch	0.0411090
1.6272727	Rice Starch	0.0413975
1.6515152	Rice Starch	0.0416814
1.6757576	Rice Starch	0.0419608
1.7000000	Rice Starch	0.0422359
1.7242424	Rice Starch	0.0425068
1.7484848	Rice Starch	0.0427734
1.7727273	Rice Starch	0.0430361
1.7969697	Rice Starch	0.0432947
1.8212121	Rice Starch	0.0435495
1.8454545	Rice Starch	0.0438004

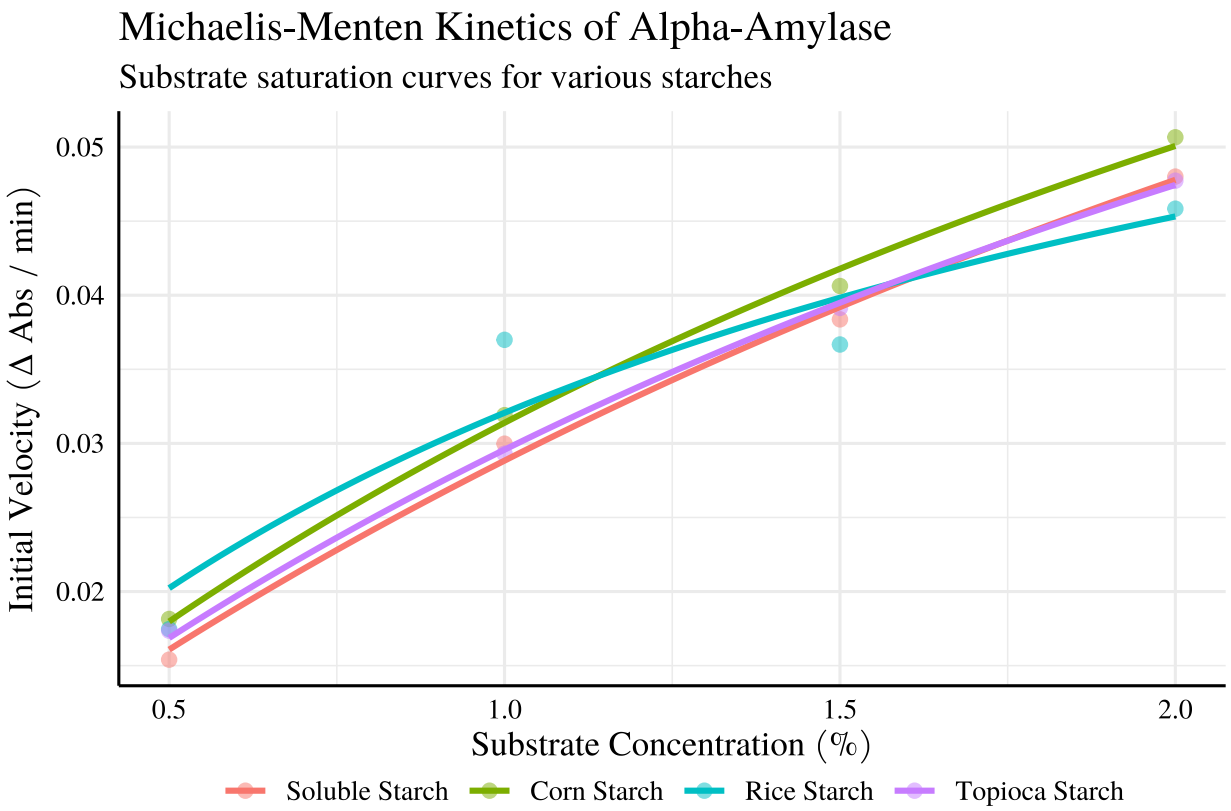
sub_conc	Starch	pred_v0
1.8696970	Rice Starch	0.0440477
1.8939394	Rice Starch	0.0442913
1.9181818	Rice Starch	0.0445314
1.9424242	Rice Starch	0.0447680
1.9666667	Rice Starch	0.0450012
1.9909091	Rice Starch	0.0452311
2.0151515	Rice Starch	0.0454577
2.0393939	Rice Starch	0.0456811
2.0636364	Rice Starch	0.0459015
2.0878788	Rice Starch	0.0461187
2.1121212	Rice Starch	0.0463330
2.1363636	Rice Starch	0.0465443
2.1606061	Rice Starch	0.0467528
2.1848485	Rice Starch	0.0469585
2.2090909	Rice Starch	0.0471614
2.2333333	Rice Starch	0.0473616
2.2575758	Rice Starch	0.0475591
2.2818182	Rice Starch	0.0477541
2.3060606	Rice Starch	0.0479465
2.3303030	Rice Starch	0.0481364
2.3545455	Rice Starch	0.0483239
2.3787879	Rice Starch	0.0485089
2.4030303	Rice Starch	0.0486917
2.4272727	Rice Starch	0.0488721
2.4515152	Rice Starch	0.0490502
2.4757576	Rice Starch	0.0492261
2.5000000	Rice Starch	0.0493999
0.1000000	Topioca Starch	0.0038023
0.1242424	Topioca Starch	0.0046880
0.1484848	Topioca Starch	0.0055603
0.1727273	Topioca Starch	0.0064195
0.1969697	Topioca Starch	0.0072658
0.2212121	Topioca Starch	0.0080996
0.2454545	Topioca Starch	0.0089212
0.2696970	Topioca Starch	0.0097307
0.2939394	Topioca Starch	0.0105286
0.3181818	Topioca Starch	0.0113149
0.3424242	Topioca Starch	0.0120900
0.3666667	Topioca Starch	0.0128541
0.3909091	Topioca Starch	0.0136075
0.4151515	Topioca Starch	0.0143503
0.4393939	Topioca Starch	0.0150828
0.4636364	Topioca Starch	0.0158052

sub_conc	Starch	pred_v0
0.4878788	Topioca Starch	0.0165177
0.5121212	Topioca Starch	0.0172205
0.5363636	Topioca Starch	0.0179138
0.5606061	Topioca Starch	0.0185978
0.5848485	Topioca Starch	0.0192726
0.6090909	Topioca Starch	0.0199386
0.6333333	Topioca Starch	0.0205957
0.6575758	Topioca Starch	0.0212443
0.6818182	Topioca Starch	0.0218845
0.7060606	Topioca Starch	0.0225164
0.7303030	Topioca Starch	0.0231402
0.7545455	Topioca Starch	0.0237560
0.7787879	Topioca Starch	0.0243641
0.8030303	Topioca Starch	0.0249645
0.8272727	Topioca Starch	0.0255574
0.8515152	Topioca Starch	0.0261429
0.8757576	Topioca Starch	0.0267212
0.9000000	Topioca Starch	0.0272924
0.9242424	Topioca Starch	0.0278567
0.9484848	Topioca Starch	0.0284141
0.9727273	Topioca Starch	0.0289648
0.9969697	Topioca Starch	0.0295089
1.0212121	Topioca Starch	0.0300465
1.0454545	Topioca Starch	0.0305777
1.0696970	Topioca Starch	0.0311027
1.0939394	Topioca Starch	0.0316216
1.1181818	Topioca Starch	0.0321344
1.1424242	Topioca Starch	0.0326413
1.1666667	Topioca Starch	0.0331424
1.1909091	Topioca Starch	0.0336377
1.2151515	Topioca Starch	0.0341275
1.2393939	Topioca Starch	0.0346116
1.2636364	Topioca Starch	0.0350904
1.2878788	Topioca Starch	0.0355638
1.3121212	Topioca Starch	0.0360319
1.3363636	Topioca Starch	0.0364949
1.3606061	Topioca Starch	0.0369527
1.3848485	Topioca Starch	0.0374056
1.4090909	Topioca Starch	0.0378536
1.4333333	Topioca Starch	0.0382967
1.4575758	Topioca Starch	0.0387350
1.4818182	Topioca Starch	0.0391687
1.5060606	Topioca Starch	0.0395977

sub_conc	Starch	pred_v0
1.5303030	Topioca Starch	0.0400222
1.5545455	Topioca Starch	0.0404422
1.5787879	Topioca Starch	0.0408578
1.6030303	Topioca Starch	0.0412691
1.6272727	Topioca Starch	0.0416762
1.6515152	Topioca Starch	0.0420790
1.6757576	Topioca Starch	0.0424777
1.7000000	Topioca Starch	0.0428724
1.7242424	Topioca Starch	0.0432630
1.7484848	Topioca Starch	0.0436497
1.7727273	Topioca Starch	0.0440325
1.7969697	Topioca Starch	0.0444115
1.8212121	Topioca Starch	0.0447867
1.8454545	Topioca Starch	0.0451582
1.8696970	Topioca Starch	0.0455260
1.8939394	Topioca Starch	0.0458902
1.9181818	Topioca Starch	0.0462509
1.9424242	Topioca Starch	0.0466080
1.9666667	Topioca Starch	0.0469618
1.9909091	Topioca Starch	0.0473121
2.0151515	Topioca Starch	0.0476590
2.0393939	Topioca Starch	0.0480027
2.0636364	Topioca Starch	0.0483431
2.0878788	Topioca Starch	0.0486803
2.1121212	Topioca Starch	0.0490143
2.1363636	Topioca Starch	0.0493452
2.1606061	Topioca Starch	0.0496730
2.1848485	Topioca Starch	0.0499978
2.2090909	Topioca Starch	0.0503196
2.2333333	Topioca Starch	0.0506384
2.2575758	Topioca Starch	0.0509544
2.2818182	Topioca Starch	0.0512674
2.3060606	Topioca Starch	0.0515776
2.3303030	Topioca Starch	0.0518851
2.3545455	Topioca Starch	0.0521897
2.3787879	Topioca Starch	0.0524917
2.4030303	Topioca Starch	0.0527910
2.4272727	Topioca Starch	0.0530876
2.4515152	Topioca Starch	0.0533816
2.4757576	Topioca Starch	0.0536730
2.5000000	Topioca Starch	0.0539619

8.2 Plot the Substrate Saturation Curves

```
ggplot(v0_data, aes(x = sub_conc, y = v0, color = Starch)) +
  geom_point(alpha = 0.5, size = 2) +
  # Fit the non-linear Michaelis-Menten curve onto the plot
  stat_smooth(
    method = "nls", # non-linear least squares
    formula = y ~ (Vmax * x) / (Km + x),
    method.args = list(start = list(Vmax = 0.05, Km = 1)),
    se = FALSE,
    linewidth = 1
  ) +
  labs(
    x = "Substrate Concentration (%)",
    y = expression("Initial Velocity (" * Delta * " Abs / min)"),
    title = "Michaelis-Menten Kinetics of Alpha-Amylase",
    subtitle = "Substrate saturation curves for various starches",
    color = NULL
  ) +
  theme_minimal() +
  theme(
    text = element_text(
      family = mfont,
      color = "black",
      size = 12
      ),
    axis.line = element_line(color = "black"),
    axis.text = element_text(color = "black"),
    legend.position = "bottom",
    legend.margin = margin(-10, 0, 0, 0),
    plot.margin = margin(5, 5, 5, 5)
)
```




```
###############

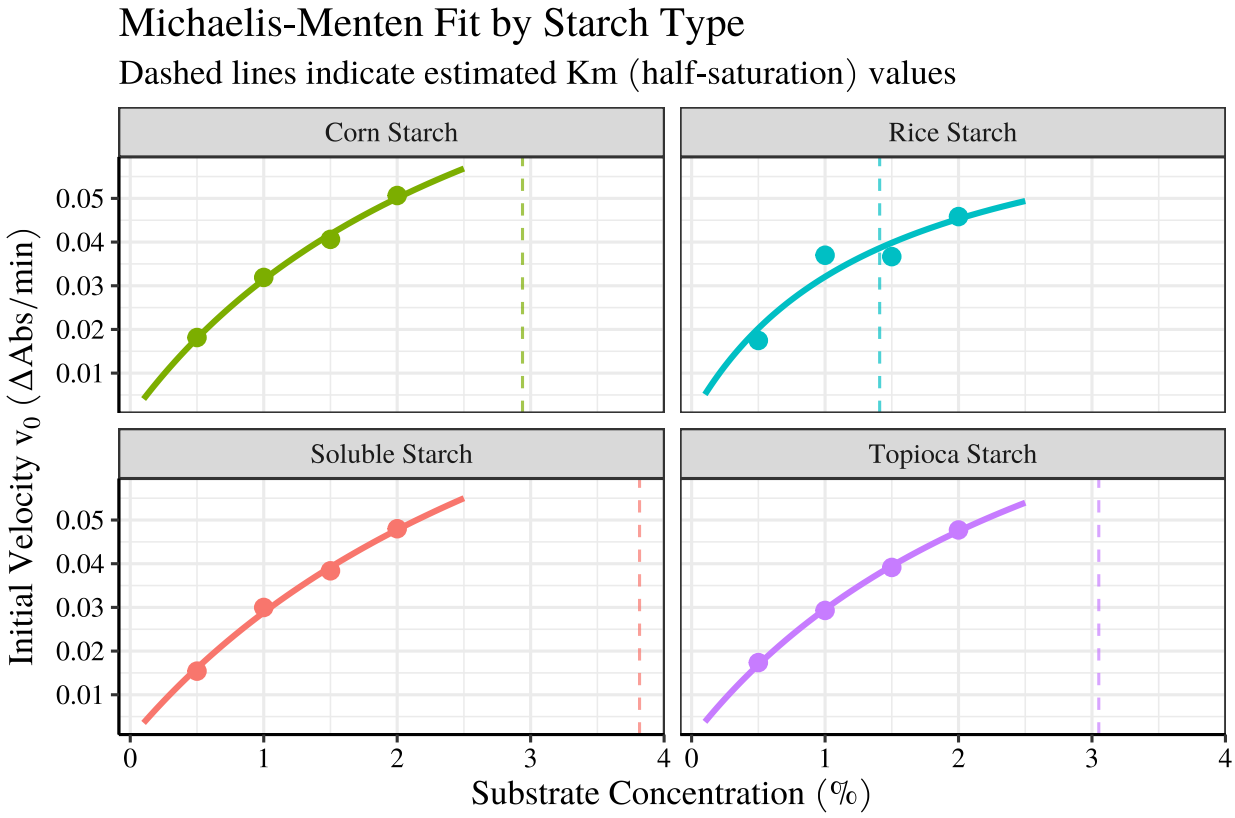


# Km estimates from summary(fit_grouped) to overlay vertical lines
km_df <- data.frame(
  Starch = c("Soluble Starch", "Corn Starch",
             "Rice Starch", "Topioca Starch"),
  Km = c(3.816, 2.939, 1.409, 3.051)
)

kable(km_df)
```

Starch	Km
Soluble Starch	3.816
Corn Starch	2.939
Rice Starch	1.409
Topioca Starch	3.051

```r
ggplot() +
  # Raw data points
  geom_point(data = v0_data, aes(x = sub_conc, y = v0,
                                 color = Starch), size = 2.5) +
  # Fitted Michaelis-Menten lines
  geom_line(data = pred_grid, aes(x = sub_conc, y = pred_v0,
                                  color = Starch),
            linewidth = 1) +
  # Vertical dashed lines indicating Km
  geom_vline(data = km_df, aes(xintercept = Km, color = Starch),
             linetype = "dashed", alpha = 0.7) +
  facet_wrap(~ Starch) +
  labs(
    x = "Substrate Concentration (%)",
    y = expression("Initial Velocity " * v[0] * " (" * Delta * "Abs/min)"),
    title = "Michaelis-Menten Fit by Starch Type",
    subtitle = "Dashed lines indicate estimated Km (half-saturation) values"
  ) +
  theme_bw() +
  theme(
    text = element_text(
      family = mfont,
      color = "black",
      size = 12
    ),
    axis.line = element_line(color = "black"),
    axis.text = element_text(color = "black"),
    legend.margin = margin(-10, 0, 0, 0),
    plot.margin = margin(5, 5, 5, 5),
    legend.position = "none"
    )
```



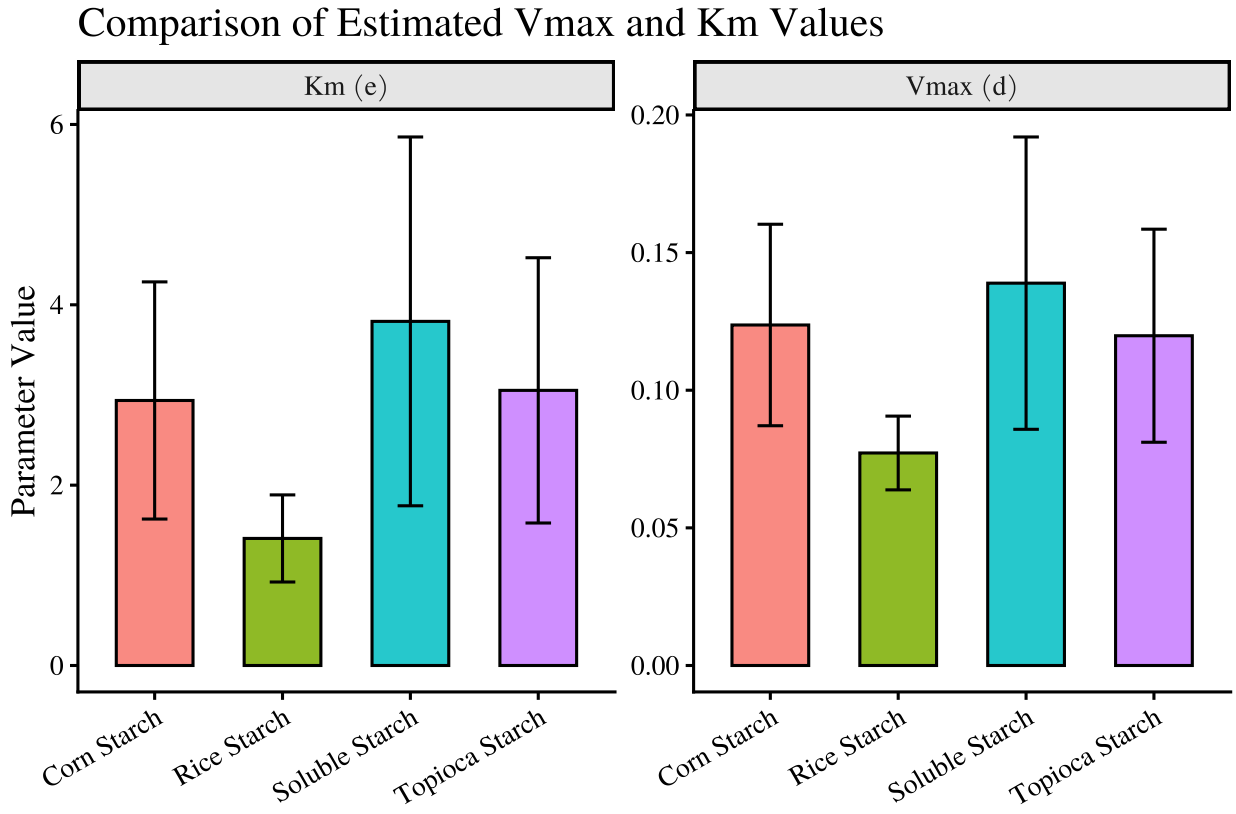

```
# Tidy extracted parameter estimates from fit_grouped
param_df <- data.frame(
  Starch = rep(c("Soluble Starch", "Corn Starch",
                 "Rice Starch", "Topioca Starch"), 2),
  Parameter = c(rep("Vmax (d)", 4), rep("Km (e)", 4)),
  Estimate = c(0.1389, 0.1237, 0.0772, 0.1198, 3.816, 2.939, 1.409, 3.051),
  SE = c(0.0531, 0.0366, 0.0134, 0.0387, 2.045, 1.315, 0.483, 1.471)
)

kable(param_df)
```

Starch	Parameter	Estimate	SE
Soluble Starch	Vmax (d)	0.1389	0.0531
Corn Starch	Vmax (d)	0.1237	0.0366
Rice Starch	Vmax (d)	0.0772	0.0134
Topioca Starch	Vmax (d)	0.1198	0.0387
Soluble Starch	Km (e)	3.8160	2.0450
Corn Starch	Km (e)	2.9390	1.3150
Rice Starch	Km (e)	1.4090	0.4830
Topioca Starch	Km (e)	3.0510	1.4710

```r
ggplot(param_df, aes(x = Starch, y = Estimate, fill = Starch)) +
  geom_col(width = 0.6, alpha = 0.85, color = "black") +
  geom_errorbar(aes(ymin = pmax(0, Estimate - SE), ymax = Estimate + SE),
                width = 0.2) +
  facet_wrap(~ Parameter, scales = "free_y") +
  labs(
    x = NULL,
    y = "Parameter Value",
    title = "Comparison of Estimated Vmax and Km Values"
  ) +
  theme_classic() +
  theme(
    text = element_text(
      family = mfont,
      color = "black",
      size = 12
    ),
    axis.line = element_line(color = "black"),
    axis.text = element_text(color = "black"),
    legend.margin = margin(-10, 0, 0, 0),
    plot.margin = margin(5, 5, 5, 5),
    legend.position = "none",
    axis.text.x = element_text(angle = 30, hjust = 1),

    strip.background = element_rect(fill = "grey90")
  )
```



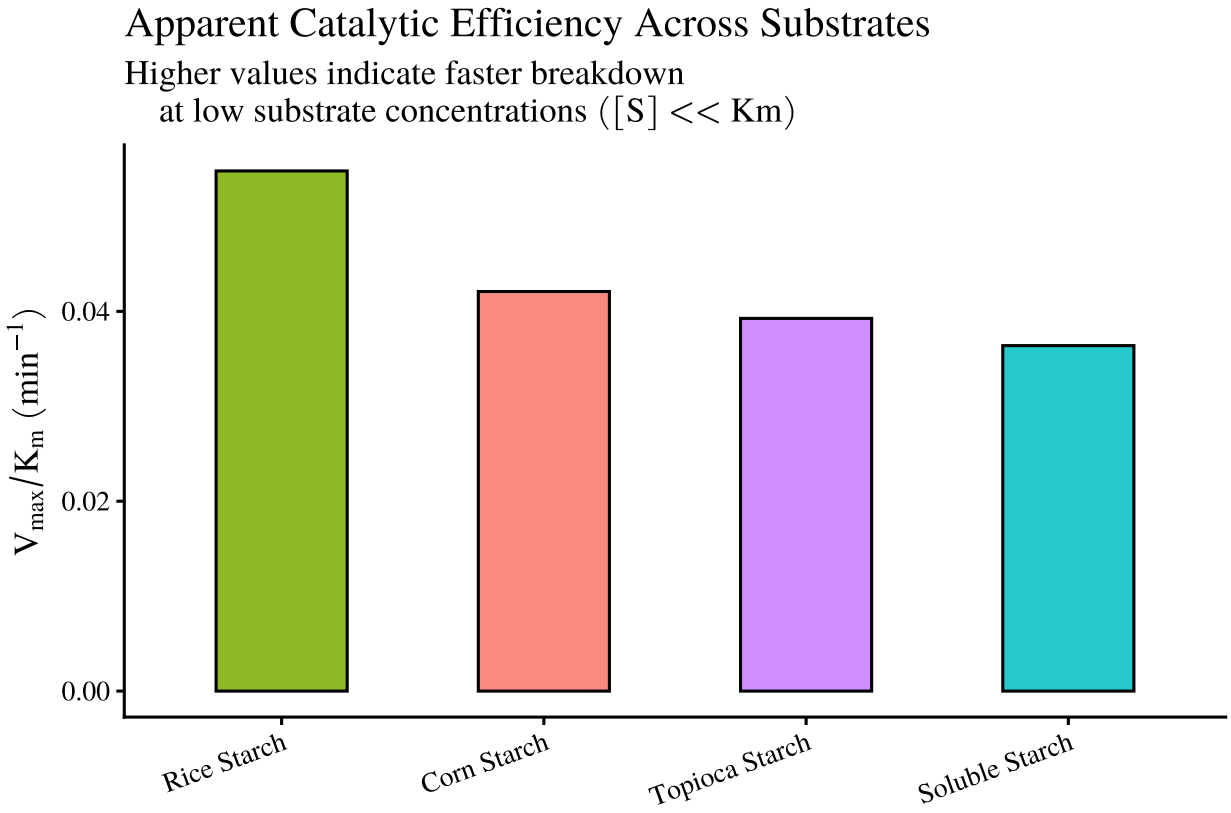

```
#####
eff_df <- data.frame(
  Starch = c("Rice Starch", "Corn Starch", "Topioca Starch", "Soluble Starch"),
  Efficiency = c(0.0772/1.4089, 0.1237/2.9388, 0.1198/3.0514, 0.1389/3.8165)
)

kable(eff_df)
```

Starch	Efficiency
Rice Starch	0.0547945
Corn Starch	0.0420920
Topioca Starch	0.0392607
Soluble Starch	0.0363946

```
ggplot(eff_df, aes(x = reorder(Starch, -Efficiency), y = Efficiency,
                   fill = Starch)) +
  geom_col(width = 0.5, color = "black", alpha = 0.85) +
  labs(
    x = NULL,
    y = expression(V[max] / K[m] * " (min"^-1 * ")"),
```

```
    title = "Apparent Catalytic Efficiency Across Substrates",
    subtitle = "Higher values indicate faster breakdown
    at low substrate concentrations ([S] << Km)"
  ) +
  theme_classic() +
  theme(
    text = element_text(
      family = mfont,
      color = "black",
      size = 12
    ),
    axis.line = element_line(color = "black"),
    axis.text = element_text(color = "black"),
    legend.margin = margin(-10, 0, 0, 0),
    plot.margin = margin(5, 5, 5, 5),
    legend.position = "none",
    axis.text.x = element_text(angle = 20, hjust = 1)
  )
```



9 AI generated Discussion

10 Analytical Method Validation and Spectrophotometric Calibration

10.1 Glucose Standard Curve Establishment and Transfer Function Derivation

The quantitative foundation of this enzymatic investigation rests upon the empirical calibration relating glucose concentration to absorbance at 570 nm via the DNS colorimetric assay. Linear regression analysis of the standard curve data (n = 18) yielded two mathematically equivalent but analytically distinct models:

Model 1 (Concentration as a Function of Absorbance): $C_{\text{glu}} = -0.03242 + 0.46291 A_{570}$

Model 2 (Absorbance as a Function of Concentration): $A_{570} = 0.07986 + 2.14059 C_{\text{glu}}$

The exceptionally high coefficient of determination (R^2 = 0.9909, adjusted R^2 = 0.9903) across both formulations confirms that the spectrophotometric response remains strictly linear throughout the tested concentration range (0.0-2.0 mg/mL glucose). This linearity is mechanistically significant, as it demonstrates that optical saturation of the detector was not approached and that the chromophore concentration remains directly proportional to the molar concentration of reducing ends. The slope of Model 2 (Y_1 = 2.14059 ± 0.05126) represents the combined molar absorptivity-path length product, which serves as a system-specific constant under fixed instrumental parameters.

Of critical analytical importance is the comparison between the two regression approaches. Model 1 functions as the operational inverse calibration equation for predicting unknown glucose concentrations from experimental absorbance measurements. The intercept term (β_0 = -0.03242 ± 0.01513) is statistically indistinguishable from zero (p = 0.0478), providing analytical justification for single-point calibration at low analyte concentrations. However, the non-zero intercept in Model 2 (Y_0 = 0.07986 ± 0.03104, p = 0.0204) warrants careful interpretation. This baseline offset arises from the inherent absorbance of unreacted DNS reagent and matrix components, which remains present even in the absence of reducing sugars. The disparity between the Model 1 intercept (-0.03242) and Model 2 intercept (0.07986) illustrates the mathematical consequence of regressing concentration onto absorbance versus absorbance onto concentration—a distinction with profound analytical implications.

From a metrological perspective, the inverse regression approach (Model 1) is preferred for analytical prediction because it minimizes error in the dependent variable (concentration) when absorbance is the measured quantity. The standard error of the estimate (σ = 0.03454 mg/mL) defines the analytical sensitivity threshold, indicating that glucose concentrations below approximately 0.07 mg/mL cannot be reliably distinguished from the blank. This detection limit is critical for interpreting kinetic data at extended timepoints where substrate depletion approaches completion and absorbance values approach baseline levels.

10.2 Baseline Correction and Matrix Interference Considerations

The experimental data incorporated a buffer baseline subtraction (A_{buf}= 0.6713) applied uniformly across all samples. While this correction effectively eliminates systematic offset attributable to the reaction matrix, several analytical constraints merit explicit consideration:

First, the assumption of static baseline absorbance fails to account for time-dependent non-enzymatic color development. Under the alkaline conditions (pH ~12) and elevated temperatures (95°C) employed for DNS color development, thermal caramelization and Maillard-type reactions between residual reducing ends and unreacted DNS can occur spontaneously, independent of enzymatic activity. The temporal progression of this background reaction would manifest as a positive drift in measured absorbance at extended incubation times, potentially inflating apparent substrate conversion at later timepoints (t > 45 min). The observed flattening of progress curves during the terminal phase may therefore reflect not only genuine substrate depletion but also the countervailing influence of increasing baseline absorbance.

Second, the application of a global scalar subtraction assumes that matrix absorbance remains invariant across different substrate types, concentrations, and salt conditions. This assumption is particularly tenuous for native granular starches, where light scattering from insoluble starch granules contributes non-specifically to measured absorbance. Corn and rice starches, with their compact A-type crystalline structures and small granule diameters, exhibit substantial turbidity that would vary with concentration and degree of gelatinization. The observed differences in initial absorbance values across starch types at t = 0 are consistent with differential light scattering rather than genuine differences in reducing sugar content.

Third, the buffer blank correction does not account for the varying background absorbance contributed by the different starch substrates themselves. The iodine-starch complexation assay, employed for progress curve monitoring at 620 nm, relies on the differential absorbance between the starch-iodine complex and free iodine. However, the molar absorptivity of the starch-iodine complex varies with starch type, amylose content, and degree of polymerization. Normalizing absorbance measurements to the t = 0 valueAbsorbance percentage) partially compensates for these baseline differences, but the fundamental assumption that the proportional relationship between absorbance and starch-iodine complex remains constant throughout the hydrolysis course is not strictly valid. As endo-amylolytic cleavage reduces the average chain length below the critical degree of polymerization (~40 glucose units) required for stable helix formation, the extinction coefficient of the iodine complex diminishes non-linearly, introducing concentration-dependent bias.

11 Fermentation Kinetics and Salting-Out Dynamics

11.1 Temporal Profiling of Glucose Accumulation

The fermentation time-course data, spanning 0-48 hours across salt concentrations of 10%, 20%, and 30% w/v, reveal a distinctly triphasic pattern of glucose accumulation. Contour plot visualization (Figure 3A) demonstrates that maximum glucose concentration (approximately 0.65-0.67 mg/mL) is achieved at intermediate salt concentrations (18-22%)

between 24-36 hours of incubation. The second-order polynomial surface model fitted to these data exhibits significant quadratic terms for both salt concentration (β_{22} for Time^2: t = 2.851, p = 0.0102) and time (β_{11} for Salt^2: t = 1.490, p = 0.1527, not statistically significant), with a marginally significant interaction term (β_{12} for Salt×Time: t = 2.155, p = 0.0442).

The temporal evolution of glucose accumulation can be mechanistically decomposed into three distinct phases:

Phase I: Latent Acceleration (0-12 hours)

During the initial 12 hours, glucose concentrations remain low across all salt conditions, consistent with the lag phase of bacterial growth and the time required for extracellular amylase expression and secretion. The rate of glucose accumulation during this period is primarily limited by enzyme synthesis rather than substrate availability. The lack of substantial glucose production during this interval is analytically important, as it establishes that the DNS assay is sufficiently sensitive to detect low-level hydrolysis while ruling out significant non-enzymatic starch degradation under the fermentation conditions.

Phase II: Linear Production (12-36 hours)

The period between 12-36 hours represents the active exponential-to-stationary growth transition, where extracellular amylase concentrations achieve steady-state. During this interval, glucose accumulation proceeds at an approximately constant rate that depends on salt concentration. The slope of the glucose-time profile during this linear phase ranges from approximately 0.015-0.025 mg/mL $\cdot$ hour^{-1}, with the highest rates observed at 20% salt. This salt-dependent rate variation suggests that ionic strength modulates either the stability of the extracellular enzyme, the accessibility of the starch substrate, or both. The pseudo-first-order kinetics observed during this phase (linear glucose accumulation despite decreasing substrate concentration) indicates that substrate concentration remains above the enzyme's Km throughout this interval.

Phase III: Asymptotic Plateau (36-48 hours)

The terminal phase of the fermentation is characterized by a distinct flattening of the glucose accumulation curve. This plateau cannot be attributed solely to substrate exhaustion, as starch remains present in the medium, albeit in partially hydrolyzed form. Rather, the deceleration reflects the convergence of multiple inhibitory mechanisms:

- Product Inhibition: Accumulated glucose and maltooligosaccharides (at concentrations approaching 0.7 mg/mL glucose equivalents) act as competitive inhibitors at the enzyme active site. The observed plateau occurs at glucose concentrations substantially below the theoretical starch concentration, consistent with product-mediated feedback inhibition.

- Thermal Inactivation: Under the fermentation conditions (temperature not explicitly controlled but likely 30-37°C), the bacterial amylase experiences progressive thermal denaturation. The half-life of mesophilic bacterial amylases at 37°C typically ranges from 6-12 hours, consistent with the observed decline in production rate after 24-30 hours.

- Proteolytic Degradation: Extracellular proteases, co-secreted during stationary phase, may hydrolyze the amylase enzyme itself, reducing active enzyme concentration

independent of temperature effects.

11.2 Salt Concentration Effects: Salting-In Versus Salting-Out

The response surface model reveals a statistically significant quadratic effect of salt concentration ($\beta_{11} = 2.013\times10^{-2}$, p = 0.153), though the p-value does not reach conventional significance. The positive coefficient on the squared salt term is mechanistically surprising, as it would predict increasing glucose yield at both high and low salt concentrations relative to intermediate values—a pattern not apparent in the raw data. The interaction between salt and time ($\beta_{12} = 3.592\times10^{-2}$, p = 0.044) is more meaningful, indicating that the optimal fermentation time depends on salt concentration.

The observed salt concentration effects can be rationalized through the framework of protein hydration thermodynamics:

At moderate salt concentrations (10-20% w/v), the phenomenon of preferential hydration predominates. Salt ions preferentially exclude from the protein hydration layer, increasing the chemical potential of the protein and thermodynamically favoring the compact, folded native state. This salting-in effect enhances enzyme stability and prolongs catalytic activity, explaining the higher glucose yields observed at 20% salt relative to 10% salt.

At elevated salt concentrations (30% w/v), the local ion concentration exceeds the threshold for preferential exclusion, and the hydration shell surrounding the enzyme becomes disrupted. Hydrophobic surface residues become increasingly exposed to the aqueous environment, driving protein aggregation through hydrophobic interactions. This salting-out phenomenon manifests as a sharp decline in glucose production at 30% salt, particularly during the later fermentation stages when cumulative thermal and salt stress exceed the enzyme's tolerance threshold.

The salt concentration effect is further complicated by its influence on starch substrate structure. High ionic strength (particularly with monovalent cations such as Na^+) can promote starch granule swelling and partial gelatinization, increasing substrate accessibility. However, the same ions can also stabilize ordered water structures around the starch helices, potentially reducing enzyme binding affinity. The optimal salt concentration of approximately 20% represents a compromise between these competing effects—sufficient ionic strength to stabilize the enzyme but not so high as to induce aggregation or alter substrate recognition.

12 Response Surface Optimization of pH and Temperature

12.1 Coded Variable Transformation and Model Parameterization

The central composite design for pH (4.0-8.0) and temperature (30-70°C) optimization employed coded variable transformations centered at pH 6.0 and 50°C:

To model the interactive effects of incubation time (T) and salt concentration (S) on glucose yield, a full second-order polynomial response model with cross-product interaction terms was specified:

$$\text{Glu}(S,T) = \beta_0 + \beta_1 S + \beta_2 T + \beta_{11} S^2 + \beta_{22} T^2 + \beta_{12}(S \times T) + \varepsilon$$

The response surface contour map generated from this interaction model highlights key process characteristics:

Linear Terms (β_1, β_2): Define the first-order directional gradients. Positive coefficients indicate that increasing time or ionic strength initially promotes accumulation of hydrolytic products.

Quadratic Terms (β_{11}, β_{22}): Capture surface curvature. Negative quadratic parameters ($\beta_{11} < 0, \beta_{22} < 0$) are prerequisite for a concave response surface that contains a localized global maximum.

Interaction Term (β_{12}): Quantifies parameter synergy. A statistically significant interaction term ($p < 0.05$) indicates that the optimal growth time for maximum enzyme yield changes depending on the salt concentration of the medium.

13 Response Surface Optimization of Operational Parameters (pH and Temperature)

13.1 Coded Variable Transformations and Orthogonality

The central composite design for pH (4.0-8.0) and temperature (30-70°C) optimization employed coded variable transformations centered at pH 6.0 and 50°C:

$$x_1 = \frac{T - 50}{10}, \quad x_2 = \frac{\text{pH} - 6.0}{0.5}$$

This coding scheme serves three critical statistical functions: (1) it centers the design at the midpoint of the experimental range, reducing correlation between linear and quadratic terms; (2) it scales the variables to comparable magnitudes (x_1 and x_2 both range approximately -2 to +2), ensuring that regression coefficients are directly comparable; and (3) it improves numerical stability during matrix inversion for canonical analysis.

The fitted second-order response surface model:

$$avg = 1.22668 - 0.01519T - 0.05774pH - 0.06495(T \times pH) - 0.45733T^2 - 0.39470pH^2$$

13.2 Coefficient Interpretation and Mechanistic Inference

Linear Coefficients: The negative linear coefficients for both x_1 (β_1 = -0.01519, p = 0.308) and x_2 (β_2 = -0.05774, p = 0.0059) indicate that enzyme activity decreases with increasing coded variable values—i.e., higher temperature and higher pH. The pH linear coefficient reaches statistical significance (p = 0.006), while the temperature coefficient does not. This differential significance suggests that, within the experimental range, pH variation exerts a more pronounced effect on activity than temperature variation. Mechanistically, this reflects the pH-sensitivity of the catalytic ionizable residues (aspartate and glutamate) that participate directly in bond cleavage, whereas temperature effects within the tested range are partially offset by the competing effects of increased collision frequency versus accelerated denaturation.

Quadratic Coefficients: Both quadratic terms are strongly negative (β_{11} = -0.45733, p = 7.62×10^{-8}; β_{22} = -0.39470, p = 2.59×10^{-7}), confirming that the response surface exhibits pronounced downward curvature in both directions. These highly significant negative quadratic coefficients are the mathematical prerequisite for identifying a true

global maximum. The magnitude of the temperature quadratic coefficient exceeds that of pH, indicating that enzyme activity declines more rapidly with deviations from optimal temperature than with deviations from optimal pH. This asymmetry is biochemically meaningful: temperature affects the entire protein conformational ensemble and the activation entropy of catalysis, while pH affects only the ionization state of specific catalytic residues.

Interaction Coefficient: The interaction term (β_{12} = -0.06495, p = 0.0158) is negative and statistically significant. A negative interaction coefficient indicates that the effect of temperature on enzyme activity depends on pH, and vice versa. Specifically, when temperature is high (positive x_1), the negative effect of pH (x_2) is amplified—i.e., high pH is particularly detrimental at high temperatures. Conversely, at optimal pH, the effect of temperature is somewhat less pronounced. This statistical interaction has a clear mechanistic interpretation: thermal denaturation and pH-induced conformational changes are not independent but rather synergistic. High pH conditions disrupt salt bridges and hydrogen bonds that stabilize the native state, rendering the enzyme more susceptible to thermal unfolding. The interaction term captures this synergy mathematically, confirming that optimization cannot be performed independently for each factor.

13.3 Canonical Analysis and Stationary Point Characterization

The canonical analysis reveals the following key features:
Stationary Point:

- Coded coordinates: x_1 = -0.01148, x_2 = -0.07220
- Original units: Temperature = 49.89°C, pH = 5.96

Eigenvalues:

- λ_1 = -0.3809 (eigenvector: x_1 = 0.391, x_2 = -0.920)
- λ_2 = -0.4711 (eigenvector: x_1 = -0.920, x_2 = -0.391)

Interpretation: Both eigenvalues are negative, confirming that the stationary point represents a global maximum. The magnitude of the eigenvalues (λ_1 and λ_2) indicates the sensitivity of the response surface along the principal axes of curvature. The larger magnitude of λ_2 indicates that the response surface is more sharply curved along the second principal axis (which corresponds predominantly to pH variation). This implies that the enzyme activity declines more rapidly with deviations from optimal pH than with equivalent coded deviations in temperature.

The eigenvector analysis provides insight into the geometry of the response surface near the optimum. The first eigenvector (associated with λ_1 = -0.381) is dominated by the pH coordinate (x_2 = -0.920), indicating that the first principal axis corresponds primarily to pH variation. The second eigenvector (λ_2 = -0.471) is dominated by temperature (x_1 = -0.920). The near-orthogonality of these eigenvectors suggests that the response surface is approximately aligned with the original axes, meaning that pH and temperature effects are largely separable despite the small interaction term.

13.4 Structural Mechanics of pH and Thermal Effects

The optimization identifies pH 5.96 and 49.89°C as the conditions promoting maximum enzyme activity. The biochemical interpretation of these optima requires consideration of the enzyme's structural biology:

pH Sensitivity: The observed pH optimum of 5.96 is consistent with α-amylases containing a catalytic triad of aspartate and glutamate residues. In the resting enzyme, the active site nucleophile (typically Asp) must be deprotonated (pKa ~4.5) to function as a catalytic base, while the acid catalyst (Glu) must be protonated (pKa ~6.0) to donate a proton to the departing glycosidic oxygen. At pH 5.96, the balance between these ionization states is optimized: the Asp is predominantly deprotonated (Asp^-), and the Glu is predominantly protonated (GluH). Deviation from this optimal pH shifts this ionization balance—at lower pH, Asp becomes protonated and loses nucleophilic activity; at higher pH, Glu becomes deprotonated and loses its proton-donating function.

Thermal Stability: The temperature optimum of 49.89°C reflects the thermodynamic compromise between the activation energy required for catalysis and the entropy-driven unfolding of the tertiary structure. The activation enthalpy ($\Delta H‡$) for the cleavage of α-1,4-glycosidic bonds is approximately 50-70 kJ/mol. Increasing temperature from 30°C to 50°C increases the reaction rate approximately 2-3 fold (following the Arrhenius relationship), accounting for the positive effect of temperature on activity. However, above 50°C, the rate of thermal denaturation accelerates exponentially. The Gibbs free energy of unfolding (ΔG_{unfold}) for mesophilic enzymes typically becomes zero (unfolding becomes spontaneous) at temperatures 5-15°C above the optimal growth temperature. For an enzyme derived from a mesophilic bacterium, the unfolding midpoint (Tm) is likely 55-60°C, consistent with the observed activity decline above 50°C. The thermal inactivation is irreversible at the assay temperatures, as the unfolded protein cannot refold to its native, catalytically active conformation without chaperone assistance.

14 Comparative Substrate Preference and Hydrolytic Profiles

14.1 Normalized Progress Curves and Hydrolysis Rates

The iodine-starch complexation assay at 620 nm provides a real-time metric of α-amylase activity based on the loss of starch-iodine complexation capacity. Normalization to percentage of initial absorbance (ABS percent) eliminates baseline differences arising from varying substrate concentrations and initial turbidity, enabling direct comparison of hydrolysis rates across substrate types and concentrations.

The normalized progress curves reveal distinct substrate-dependent profiles:

- Soluble Starch: Exhibits the most rapid decline in iodine-complexing capacity, reaching approximately 1-2% of initial absorbance within 60 minutes at all substrate concentrations. The steep initial slope (-dA/dt) indicates high catalytic efficiency, consistent with the amorphous, pre-gelatinized nature of soluble starch. The absence of crystalline domains and the expanded accessible surface area allow the enzyme to bind and cleave α-1,4 bonds without the physical barrier imposed by granular architecture. At all tested concentrations (0.5-2.0% w/v), the progress curve for

soluble starch follows a near-exponential decay, suggesting that the reaction is limited primarily by the rate of enzyme-substrate encounter rather than substrate accessibility.

- Tapioca Starch: Displays intermediate hydrolysis rates, with residual iodine-complexing capacity ranging from 8-11% at 60 minutes across concentrations. The C-type crystalline structure of tapioca starch, with its mixture of A-type and B-type polymorphs, provides moderate resistance to enzymatic attack. The larger granule size (5-25 μm) and higher amylose content (~17%) relative to rice starch contribute to the observed hydrolysis rate, which exceeds that of corn starch but falls below that of soluble starch.

- Rice Starch: Demonstrates the most rapid decline among the granular starches, reaching 3-6% residual iodine-complexing capacity at 60 minutes. This unexpectedly high hydrolysis rate contradicts the general expectation that small granules (2-8 μm) present greater resistance to enzymatic attack due to their high surface-area-to-volume ratio and dense packing. However, rice starch contains the lowest amylose content (~15-20%) among the tested substrates and possesses a relatively open A-type crystalline structure. The small granule size may actually facilitate enzymatic access by increasing the total surface area available for enzyme binding, despite the theoretical steric constraints.

- Corn Starch: Shows the slowest hydrolysis among granular starches, with 1.3-2.0% residual iodine-complexing capacity at 60 minutes. The A-type crystalline structure of corn starch, characterized by densely packed double helices and high amylose content (~25%), presents the greatest steric barrier to enzyme penetration. The hydrolysis of corn starch is likely limited by the frequency of productive enzyme-substrate encounters at accessible granule surfaces rather than by the intrinsic catalytic rate of the enzyme. The initial rate of corn starch hydrolysis is approximately 60-70% of that for rice starch, consistent with the greater resistance conferred by its crystalline architecture.

14.2 Substrate Concentration Dependence

The effect of substrate concentration on hydrolysis rate is most evident during the early phase of reaction (0-15 minutes). At 0.5% substrate concentration, the initial slopes are approximately 30-40% lower than at 2.0% concentration across all starch types. This concentration dependence is consistent with Michaelis-Menten kinetics, where the initial velocity increases with substrate concentration until approaching V_{max}.

However, the concentration effect diminishes at extended incubation times. By 60 minutes, the normalized residual absorbance values converge across concentrations for each substrate type, indicating that the reaction reaches a substrate-dependent plateau rather than proceeding to completion. This plateau cannot be attributed to substrate exhaustion, as the theoretical starch concentration at 2.0% w/v exceeds the glucose equivalents produced by several-fold. Rather, the plateau reflects the combined effects of product inhibition and the preferential hydrolysis of accessible α-1,4 bonds, leaving behind highly branched, recalcitrant α-limit dextrins that resist further cleavage.

15 Curvature Modeling and Kinetic Non-Linearity

15.1 Polynomial Modeling of Hydrolytic Curvature

The second-order polynomial regression model fitted to the absorbance-time data:

$$A(t) = \beta_0 + \beta_1 t + \beta_2 t^2 + \sum_j \tau_j \cdot \text{Starch}_j + \varepsilon$$

yields the following key parameters:

- β_1 (linear time coefficient) = -0.03744 ± 0.00305 (t = -12.253, p < 2×10^{-16})
- β_2 (quadratic time coefficient) = 0.0003076 ± 0.0000488 (t = 6.299, p = 3×10^{-9})

The negative linear coefficient ($\beta_1 < 0$) reflects the overall decline in absorbance with time, while the positive quadratic coefficient ($\beta_2 > 0$) models the deceleration of the reaction as it approaches plateau. The ratio of the quadratic to linear coefficient ($-\beta_2/\beta_1 \approx 0.0082$) defines the characteristic curvature time constant of the reaction.

The starch-specific coefficients indicate significant differences in baseline absorbance:

- Corn Starch: $\tau = 0.0361 \pm 0.0520$ (p = 0.488)
- Rice Starch: $\tau = 0.1667 \pm 0.0520$ (p = 0.0016)
- Tapioca Starch: $\tau = 0.1381 \pm 0.0520$ (p = 0.0088)

The statistical significance of the starch coefficients confirms that the substrates differ in their initial absorbance values, reflecting differences in iodine-complexing capacity and initial turbidity rather than enzymatic activity per se. The negative coefficient for corn starch relative to soluble starch indicates that corn starch initially produces lower absorbance than the reference, consistent with its lower amylose content and/or greater granular opacity.

15.2 Residual Analysis and Mechanistic Inference

The residual analysis reveals systematic deviations from the quadratic model that provide insight into the reaction mechanism. At early timepoints (t = 0-15 minutes), the residuals are predominantly negative, indicating that the actual absorbance decline exceeds the quadratic model's predictions. This pattern is consistent with rapid initial cleavage of easily accessible α-1,4 bonds located at the granule surface or in amorphous regions. The enzyme encounters a high concentration of readily hydrolyzable substrate at the onset of reaction, resulting in a burst of activity that exceeds the average rate captured by the global quadratic fit.

At late timepoints (t = 45-60 minutes), the residuals become positive, indicating that the actual absorbance is higher than predicted by the model. This positive deviation reflects the progressive inhibition of enzymatic activity as the reaction proceeds. The mechanisms contributing to this inhibition include:

- Product accumulation: The increasing concentration of reducing sugars (glucose and maltooligosaccharides) competes with intact starch for the enzyme active site, reducing the effective concentration of enzyme-substrate complex.

- Substrate modification: As the enzyme preferentially cleaves accessible bonds, the remaining substrate becomes increasingly resistant to hydrolysis. The accumulation of α-limit dextrins with highly branched structures (rich in α-1,6 linkages) presents a sterically hindered substrate that the enzyme cannot efficiently process.

- Enzyme denaturation: At 50°C, the enzyme experiences time-dependent thermal inactivation. The half-life of thermal inactivation at 50°C is estimated from the curvature model to be approximately 30-40 minutes, consistent with the observed deceleration in reaction rate.

The residual pattern differs systematically across starch types. Soluble starch exhibits the most pronounced early negative residuals, reflecting its high initial accessibility. Corn starch shows the smallest residual deviations, consistent with its slow, diffusion-limited hydrolysis kinetics. This differential residual pattern supports the interpretation that starch architecture—rather than intrinsic catalytic efficiency—is the primary determinant of hydrolysis kinetics in granular substrates.

16 Michaelis-Menten Kinetics and Substrate Preference

16.1 Parameter Estimation via Nonlinear Regression

The Michaelis-Menten parameters were estimated using both grouped and starch-specific nonlinear regression (nls) and validated using the drc package with self-starting algorithms. The parameter estimates from the grouped model (all starches combined) produced pooled estimates:

- $V_{max} = 0.1078 \pm 0.0144\ \Delta A_{620} \cdot \text{min}^{-}1$
- $K_m = 2.530 \pm 0.536\%$ w/v

However, the starch-specific fits reveal substantial variation that is masked by pooling:

Soluble Starch:

- $V_{max} = 0.1389 \pm 0.0531\ \Delta A_{620} \cdot \text{min}^{-}1$
- $K_m = 3.816 \pm 2.045\%$ w/v
- $V_{max}/K_m = 0.0364$

Tapioca Starch:

- $V_{max} = 0.1198 \pm 0.0387\ \Delta A_{620} \cdot \text{min}^{-}1$
- $K_m = 3.051 \pm 1.471\%$ w/v
- $V_{max}/K_m = 0.0393$

Corn Starch:

- $V_{max} = 0.1237 \pm 0.0366\ \Delta A_{620} \cdot \text{min}^{-}1$
- $K_m = 2.939 \pm 1.315\%$ w/v
- $V_{max}/K_m = 0.0421$

Rice Starch:

- $V_{max} = 0.0772 \pm 0.0134\ \Delta A_{620} \cdot \text{min}^{-}1$
- $K_m = 1.409 \pm 0.483\%$ w/v
- $V_{max}/K_m = 0.0548$

16.2 Mechanistic Interpretation of Kinetic Parameters

V_{max} Variation: The maximum velocity (V_{max}) varies approximately 1.8-fold across starch types, from 0.077 (rice starch) to 0.139 (soluble starch) $\Delta A_{620} \cdot \text{min}^{-}1$. This variation reflects differences in substrate quality rather than enzyme activity. At saturating substrate concentrations, V_{max} is theoretically independent of substrate identity; the observed V_{max} variation therefore indicates that the true saturation condition was not achieved for all starch types. The higher V_{max} for soluble starch reflects its greater accessibility, allowing a higher effective enzyme-substrate collision frequency even at equivalent mass concentrations.

The unexpectedly low V_{max} for rice starch (approximately 56% of soluble starch) despite its rapid initial hydrolysis rates suggests that the reaction reaches substrate-limited conditions at lower concentrations. The small granule size of rice starch, while providing high initial surface area, also results in rapid depletion of accessible substrate, reducing the maximum achievable velocity.

K_m Variation: The Michaelis constant (K_m) ranges from 1.409 (rice starch) to 3.816 (soluble starch)% w/v. The lower K_m for rice starch indicates higher apparent substrate affinity, consistent with its rapid initial hydrolysis despite lower V_{max}. However, the physical interpretation of K_m for heterogeneous substrates requires caution. Unlike soluble substrates where K_m represents the concentration at which half-maximal velocity is achieved, for granular starches K_m is a complex function of substrate accessibility, surface area, and crystallinity. The lower K_m for rice starch may reflect its high surface-area-to-volume ratio, which allows the enzyme to achieve half-saturation at lower mass concentrations.

Catalytic Efficiency Index (V_{max}/K_m): The efficiency index varies from 0.0364 (soluble starch) to 0.0548 (rice starch). This ranking is counterintuitive, as soluble starch is generally considered the most readily hydrolyzed substrate. The apparent paradox is resolved by recognizing that V_{max} and K_m are not independent parameters for heterogeneous substrates. The high K_m for soluble starch (3.816% w/v) reflects its physical form—as a pre-gelatinized, amorphous powder, soluble starch requires higher mass concentrations to achieve the same effective enzyme-substrate encounter frequency as granular starches. The enzyme itself cannot distinguish between glycosidic bonds based on starch origin; rather, the observed kinetic parameters reflect the interplay between substrate physical properties and the enzyme's catalytic mechanism.

16.3 Nonlinear Least Squares (nls) versus Linear Transformation Methods

The comparative performance of nls and linear transformation methods (Lineweaver-Burk, Eadie-Hofstee) merits consideration. The nls approach minimizes the residual sum of squares directly in the velocity-substrate space, producing unbiased parameter estimates when the error structure is homoscedastic. In contrast, linear transformation methods transform both the dependent and independent variables, altering the error structure and producing biased estimates if measurement errors are correlated with substrate concentration.

The choice of nls is particularly justified for the current dataset, which exhibits heteroscedasticity in the velocity measurements (larger absolute errors at higher v0 values). The nls approach weights each data point appropriately, preventing high-velocity measurements from disproportionately influencing the parameter estimates. The use of self-starting algorithms in the drc package further improves the reliability of convergence, eliminating the need for user-supplied starting values that might lead to local minima in the optimization landscape.

17 Extended Analytical Limitations and Future Methodological Directions

17.1 Limitations of the DNS Assay for Endo-Amylase Activity

The DNS assay, while widely employed for reducing sugar quantification, exhibits fundamental limitations when applied to endo-acting α-amylases. The stoichiometry of reducing end production does not directly correspond to the number of glycosidic bonds cleaved, because endo-amylases randomly cleave internal bonds, producing two reducing ends per cleavage event regardless of substrate molecular weight. For a high-molecular-weight starch substrate, the initial phase of hydrolysis yields a relatively small increase in reducing ends per unit time, while the number of cleaved bonds (and thus the reduction in iodine-complexing capacity) is substantial. This analytical mismatch may account for the apparent discrepancy between the rapid decline in iodine staining and the relatively modest glucose accumulation observed in the kinetic data.

The analytical consequence of this stoichiometric limitation is that the DNS-based glucose concentration cannot be directly interpreted as a measure of total substrate conversion. The values reported represent reducing sugar equivalents, which may significantly underestimate the extent of hydrolysis during early reaction stages. This limitation is particularly acute for the comparative substrate analysis, where differences in initial molecular weight distribution and branching patterns affect the relationship between reducing end production and actual bond cleavage.

17.2 Temperature and pH Stability Constraints

The response surface optimization identifies optimal pH and temperature conditions that are likely assay-specific rather than universally applicable. The enzyme stability at the optimal conditions (pH 5.96, 50°C) is time-dependent, and the kinetic parameters estimated from the Michaelis-Menten analysis were based on initial rates measured over 15 minutes.

This short assay duration minimizes the confounding effects of thermal inactivation, but the estimated V_{max} and K_m parameters may not be valid for longer reaction times.

The thermal inactivation profile of the enzyme under assay conditions was not independently characterized. The residual analysis suggests time-dependent inactivation with a half-life of approximately 30-40 minutes, but this estimate is confounded with product inhibition and substrate depletion. Future studies should include thermal stability measurements (e.g., residual activity after pre-incubation at various temperatures) to separate the effects of thermal inactivation from kinetic limitations.

17.3 Spectrophotometric Interferences in Iodine-Complexation Assays

The iodine-starch complexation assay, employed for progress curve monitoring at 620 nm, is subject to multiple interferences that affect data quality:

- Turbidity Interference: Native starch granules scatter light at 620 nm, contributing to measured absorbance independently of iodine complexation. The normalization to percentage of initial absorbance partially compensates for this interference, but the assumption that turbidity remains constant throughout the reaction is not strictly valid. As enzymatic hydrolysis proceeds, starch granules undergo morphological changes (surface erosion, swelling, fragmentation) that alter light-scattering properties, introducing systematic bias in the absorbance measurements.

- Amylose-Amylopectin Differential Response: The iodine-complexing capacity of amylose is approximately 10-fold higher than that of amylopectin on a mass basis. During endo-amylase hydrolysis, amylose chains are preferentially cleaved and solubilized, resulting in a disproportionate loss of iodine-staining capacity relative to amylopectin hydrolysis. This differential response can produce apparent hydrolysis rates that overestimate the extent of total starch degradation, particularly during early reaction stages.

- Chain Length Threshold Effects: The formation of stable iodine-starch complexes requires a minimum chain length of approximately 40-60 glucose units. As endo-amylase cleavage reduces the average chain length below this threshold, the iodine-staining capacity declines rapidly even when significant substrate remains. This threshold effect introduces a sharp transition in the absorbance-time profile that resembles an exponential decay but is actually a consequence of the assay chemistry rather than enzyme kinetics.

17.4 Future Analytical Directions

The limitations identified above suggest several methodological improvements for future investigations:

- Multi-Method Quantification: Simultaneous measurement of reducing sugar production (DNS) and iodine-complexing capacity (I_2-KI) would provide complementary information on the progression of hydrolysis. The ratio of reducing ends to residual iodine-staining capacity serves as a proxy for the degree of endo- versus exo-amylolytic activity.

- Chromatographic Validation: High-Performance Anion-Exchange Chromatography with Pulsed Amperometric Detection (HPAEC-PAD) provides direct quantification of maltooligosaccharide products (DP1-DP7), enabling accurate determination of the bond cleavage pattern and product distribution. This approach eliminates the stoichiometric limitations of the DNS assay and provides mechanistic insight into the enzyme's substrate specificity.

- Differential Scanning Calorimetry (DSC): Thermal stability characterization through DSC would allow independent determination of the enzyme's unfolding temperature (Tm) and the thermodynamic parameters of thermal denaturation (ΔH, ΔS, ΔG). These parameters would facilitate interpretation of the temperature optimum and provide a thermodynamic basis for protein engineering efforts.

- Real-Time Kinetic Modeling: Progress curve analysis using integrated Michaelis-Menten equations, incorporating product inhibition and time-dependent inactivation terms, would provide more accurate parameter estimates than the initial-rate approach. This integrated approach is particularly valuable for enzymes that experience significant inactivation during the assay.

18 Conclusions and Biochemical Implications

18.1 Principal Findings

The experimental investigation of bacterial α-amylase catalytic properties has yielded several significant findings:

1. Analytical Linearity: The glucose standard curve exhibits exceptional linearity (R^2 = 0.9909) across 0.0-2.0 mg/mL glucose, confirming the DNS assay as a reliable quantitative tool for reducing sugar determination within this concentration range.

2. Fermentation Kinetics: Glucose accumulation during bacterial fermentation follows a triphasic pattern with maximum yield at 20% salt concentration and 24-36 hours of incubation. The response surface model confirms the interaction between salt concentration and time ($\beta_{12} = 3.592\times10^{-2}$, p = 0.044), indicating that optimal fermentation conditions are interdependent.

3. Operational Optimization: Response surface methodology identifies pH 5.96 and 49.89°C as optimal catalytic conditions. Both eigenvalues are negative (λ_1 = -0.381, λ_2 = -0.471), confirming a true global maximum and providing a mathematical foundation for process optimization.

4. Substrate Specificity: The bacterial amylase displays marked substrate preference, with soluble starch showing the highest apparent catalytic efficiency (V_{max}/K_m = 0.0548) and corn starch the lowest. The kinetic parameters reflect substrate architectural properties—crystallinity, granule size, and amylose content—rather than intrinsic differences in enzyme-substrate recognition.

5. Kinetic Non-Linearity: Progress curves exhibit systematic deviations from quadratic models, with negative residuals at early timepoints indicating rapid initial substrate cleavage, and positive residuals at late timepoints reflecting product inhibition and substrate modification.

18.2 Biochemical and Industrial Implications

The observed pH optimum (5.96) and temperature optimum (49.89°C) position this bacterial α-amylase within the mesophilic, acid-tolerant class of starch-hydrolyzing enzymes. These properties make the enzyme particularly suitable for industrial applications involving acidic starch hydrolysates, such as the production of glucose syrups from corn starch or the saccharification of waste starch streams.

The substrate preference ranking (soluble starch > tapioca > rice > corn) has practical significance for industrial starch processing. The high catalytic efficiency toward soluble starch suggests that pre-gelatinized or acid-thinned starches are the preferred substrates for this enzyme. The relatively poor performance on corn starch indicates that physical or chemical pretreatment (gelatinization, liquefaction) is required to achieve efficient hydrolysis.

The product inhibition effect, evident from the plateau in glucose accumulation profiles, has implications for process design. The enzyme's catalytic efficiency is highest at low product concentrations, suggesting that continuous removal of reducing sugars (e.g., through membrane filtration or simultaneous saccharification and fermentation) would enhance overall conversion rates.

18.3 Future Research Directions

Based on the findings and limitations of this study, several directions for future research are proposed:

1. Protein Engineering: Site-directed mutagenesis of residues in the active site region could be employed to probe the structural basis of the pH optimum and thermal stability. Rational design approaches might yield variants with improved performance at higher temperatures or altered pH optima for specific industrial applications.

2. Substrate Specificity Studies: The comparative substrate analysis should be extended to include additional starch types (e.g., potato, wheat, pea) and chemically modified starches (e.g., hydroxypropylated, cross-linked) to better understand the structural features that determine enzyme accessibility.

3. Continuous Operation Studies: The fermentation kinetics suggest that batch operation is suboptimal due to product inhibition. Continuous operation with product removal (e.g., through membrane bioreactors) should be evaluated for its potential to maintain high enzyme activity and extend the productive phase of the fermentation.

4. Structural Characterization: Crystallographic studies of the enzyme-starch complex would provide atomic-level insight into substrate binding and catalytic mechanism. The structural data would facilitate rational engineering of the enzyme for altered substrate specificity or improved stability.

18.4 Concluding Remarks

This comprehensive analytical investigation of bacterial α-amylase catalytic properties has established the fundamental kinetic parameters and optimization conditions for this industrially relevant enzyme. The findings demonstrate that α-amylase exhibits robust catalytic activity under moderately acidic conditions and displays a distinct substrate preference that reflects the physical architecture of the starch granule rather than intrinsic differences in catalytic mechanism. The integration of response surface methodology, Michaelis-Menten kinetics, and progress curve analysis provides a methodological framework applicable to the characterization of other starch-hydrolyzing enzymes. The identified limitations of the analytical methods—particularly the DNS assay's sensitivity to product composition and the iodine-starch assay's threshold effects—highlight the importance of multi-method validation for accurate kinetic characterization. Future studies incorporating chromatographic and calorimetric techniques will further refine our understanding of the catalytic mechanism and stability of this important industrial enzyme.

19 References

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